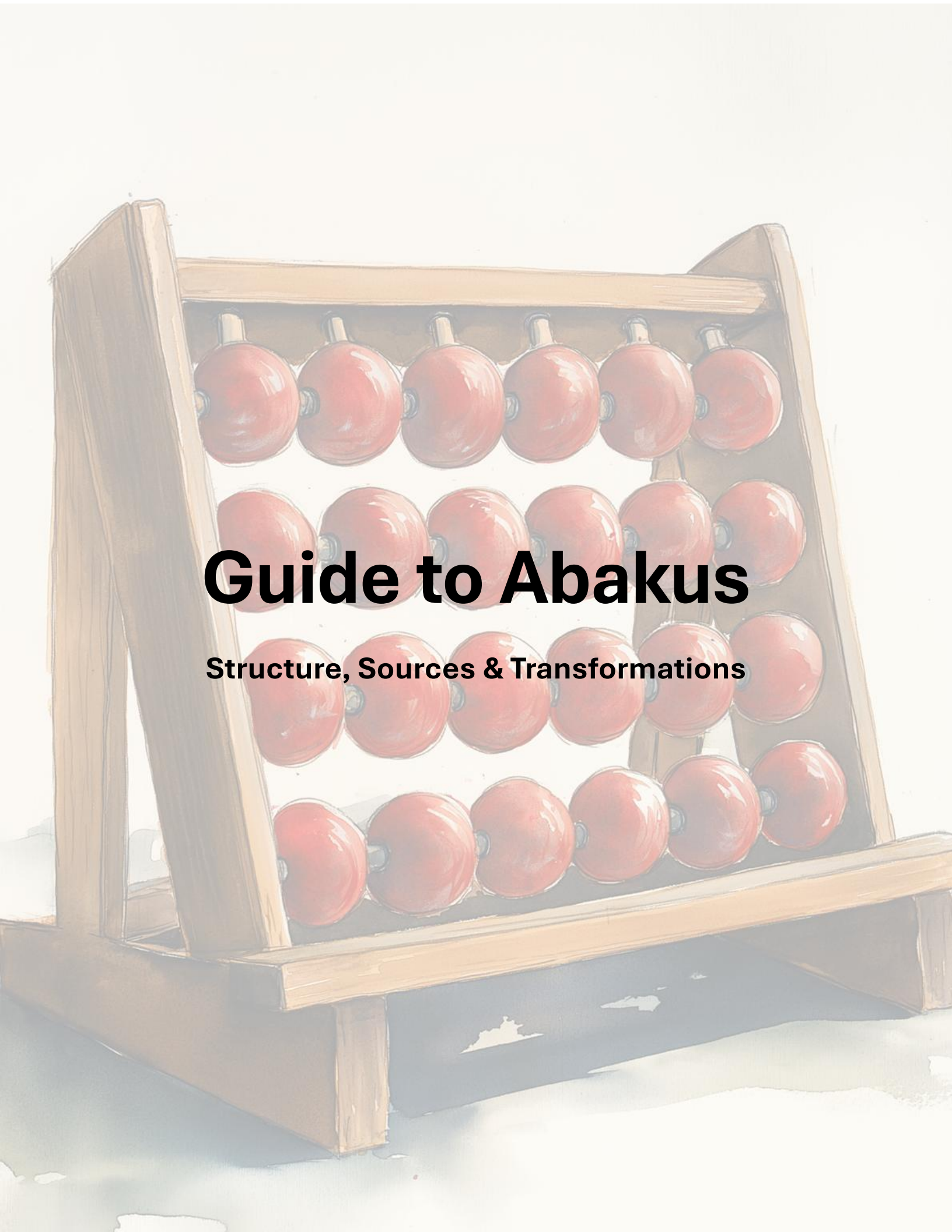
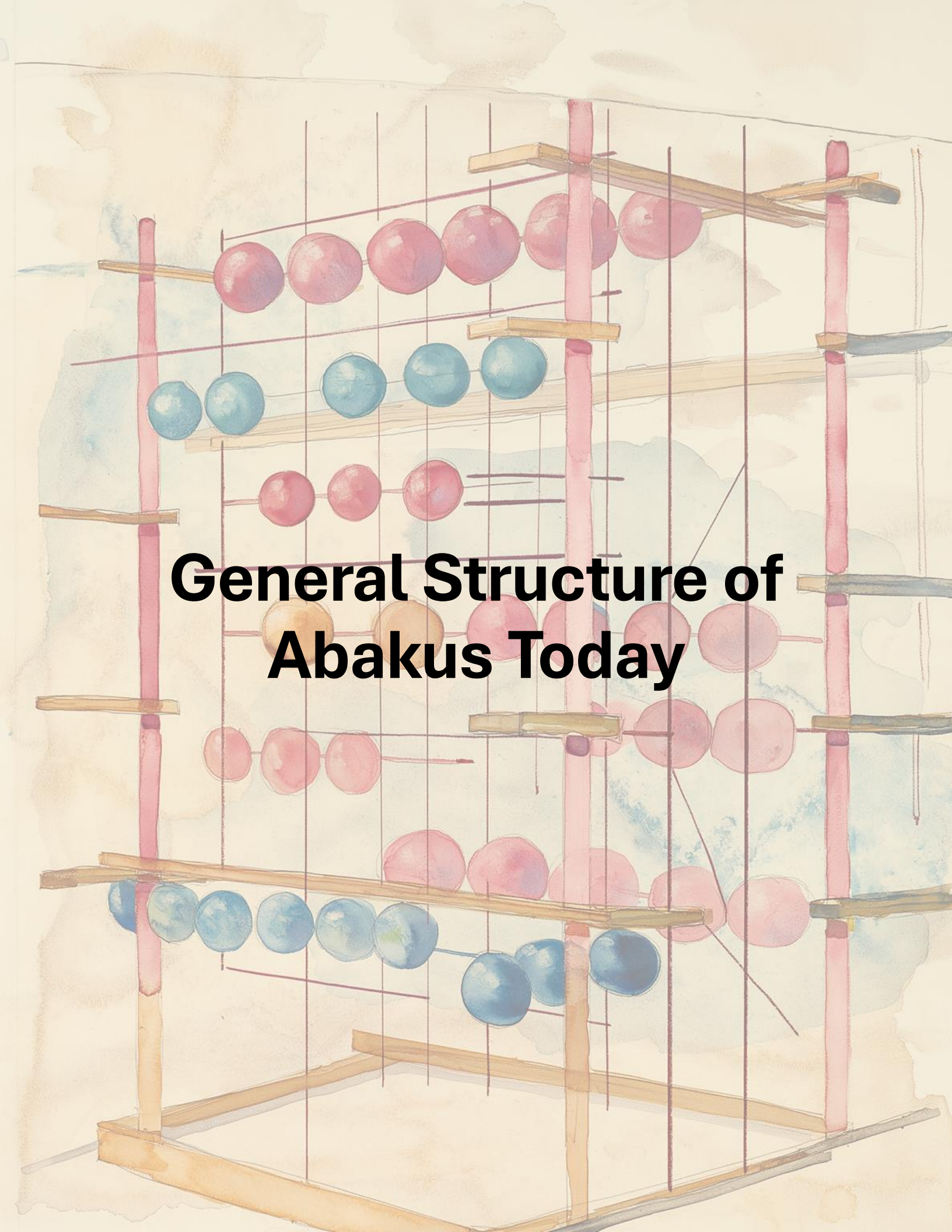




Guide to Abakus

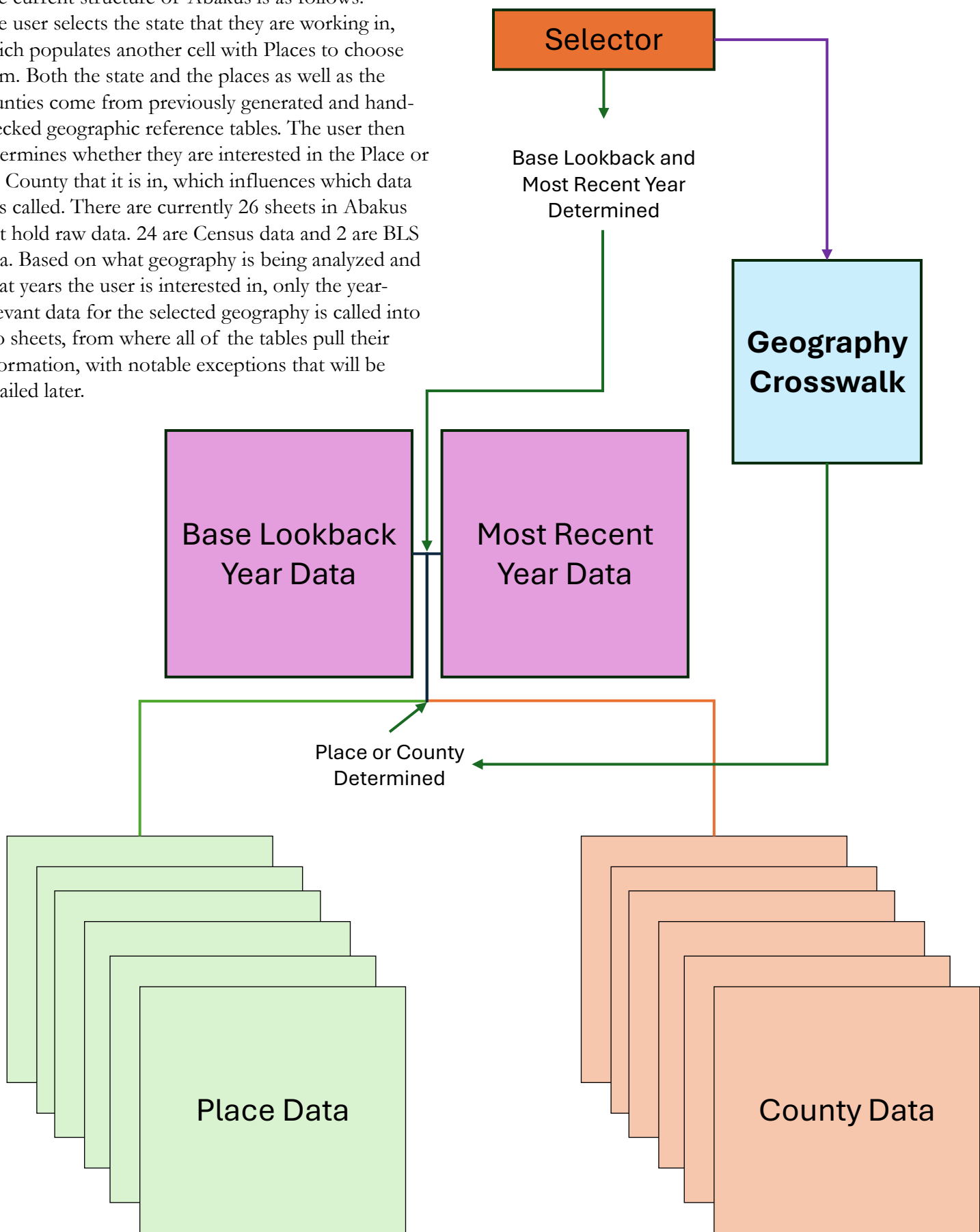
Structure, Sources & Transformations



General Structure of Abakus Today

General Structure

The current structure of Abakus is as follows:
The user selects the state that they are working in, which populates another cell with Places to choose from. Both the state and the places as well as the counties come from previously generated and hand-checked geographic reference tables. The user then determines whether they are interested in the Place or the County that it is in, which influences which data gets called. There are currently 26 sheets in Abakus that hold raw data. 24 are Census data and 2 are BLS data. Based on what geography is being analyzed and what years the user is interested in, only the year-relevant data for the selected geography is called into two sheets, from where all of the tables pull their information, with notable exceptions that will be detailed later.

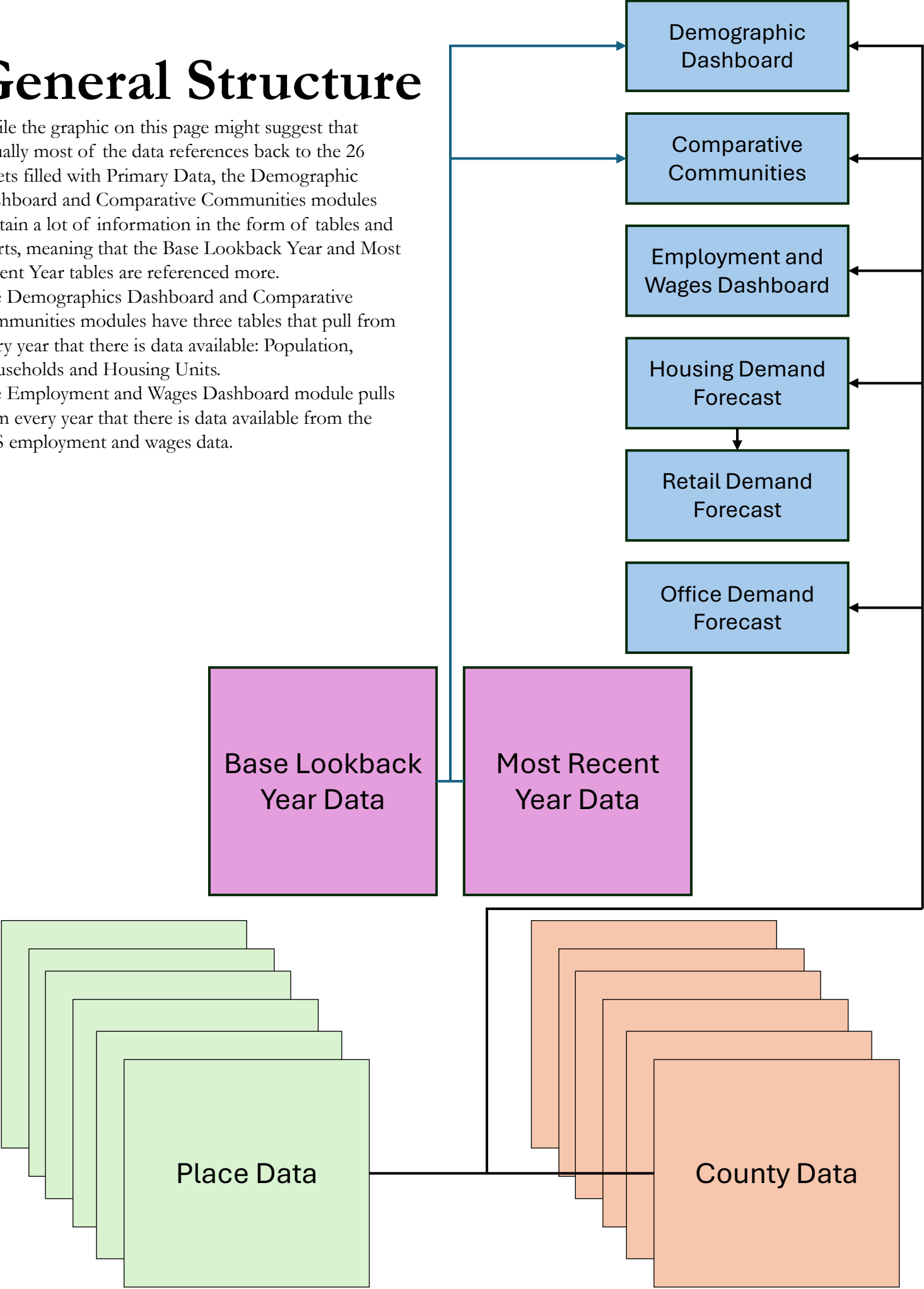


General Structure

While the graphic on this page might suggest that actually most of the data references back to the 26 sheets filled with Primary Data, the Demographic Dashboard and Comparative Communities modules contain a lot of information in the form of tables and charts, meaning that the Base Lookback Year and Most Recent Year tables are referenced more.

The Demographics Dashboard and Comparative Communities modules have three tables that pull from every year that there is data available: Population, Households and Housing Units.

The Employment and Wages Dashboard module pulls from every year that there is data available from the BLS employment and wages data.



Types of Data in Abakus

1. **Primary Data** – Primary Data is the bulk of Abakus and is defined as the raw survey data collected by Census and the Bureau of Labor Statistics. This data may be referenced to create growth ratios which may then influence the output of various demand modules.
2. **Geographic Data** – Geographic Data is data that will require yearly updating but that cannot generally be done through automated processes. These will need to be reference sheets that help drive the tool and will need to be looked over carefully before being added to the tool annually. This data provides otherwise undocumented links between Places and Counties as well as County Subdivisions which we hope to add to the tool.
3. **Secondary Reference Data** – This data differs from primary data in that it isn't pulled from surveys and the data itself is never something that gets reported out to an Abakus user. This data is built into the tool in order to inform certain outputs. The main example of this is BEA household spending data, which has to be researched, interpreted and then is input into Abakus and then once input influences the results of the Retail Demand output.
4. **Expert Assumptions** – These are figures that I am boldly claiming to be an expert on and that are programmed into Abakus. The main example of this is within the Housing Demand Forecast module I have programmed in a few different selectable options for internal turnover of households based on trial and error.

The bulk of this manual will detail the ways in which primary data is used to populate various tables and demand forecast outputs. Less time will be spent detailing the other data sources, which will be supplied as stand-alone Excel or CSV documents, which can be integrated into the tool.

A wooden arrow, light brown in color with visible wood grain, points diagonally upwards and to the right. It is positioned centrally against a dense background of green leaves, which are rendered in various shades of green with some darker outlines. The word "Selector" is printed in a bold, black, sans-serif font across the middle of the arrow.

Selector

Selector

Select Area of Study ->

Select State ↓

Select Place ↓

GA	Adairsville city, Georgia	Bartow County, Georgia
----	---------------------------	------------------------

Select Geography ->

Place	
2012	2022

Select Base and Most Recent Year ->

Select Comparison Cities ->

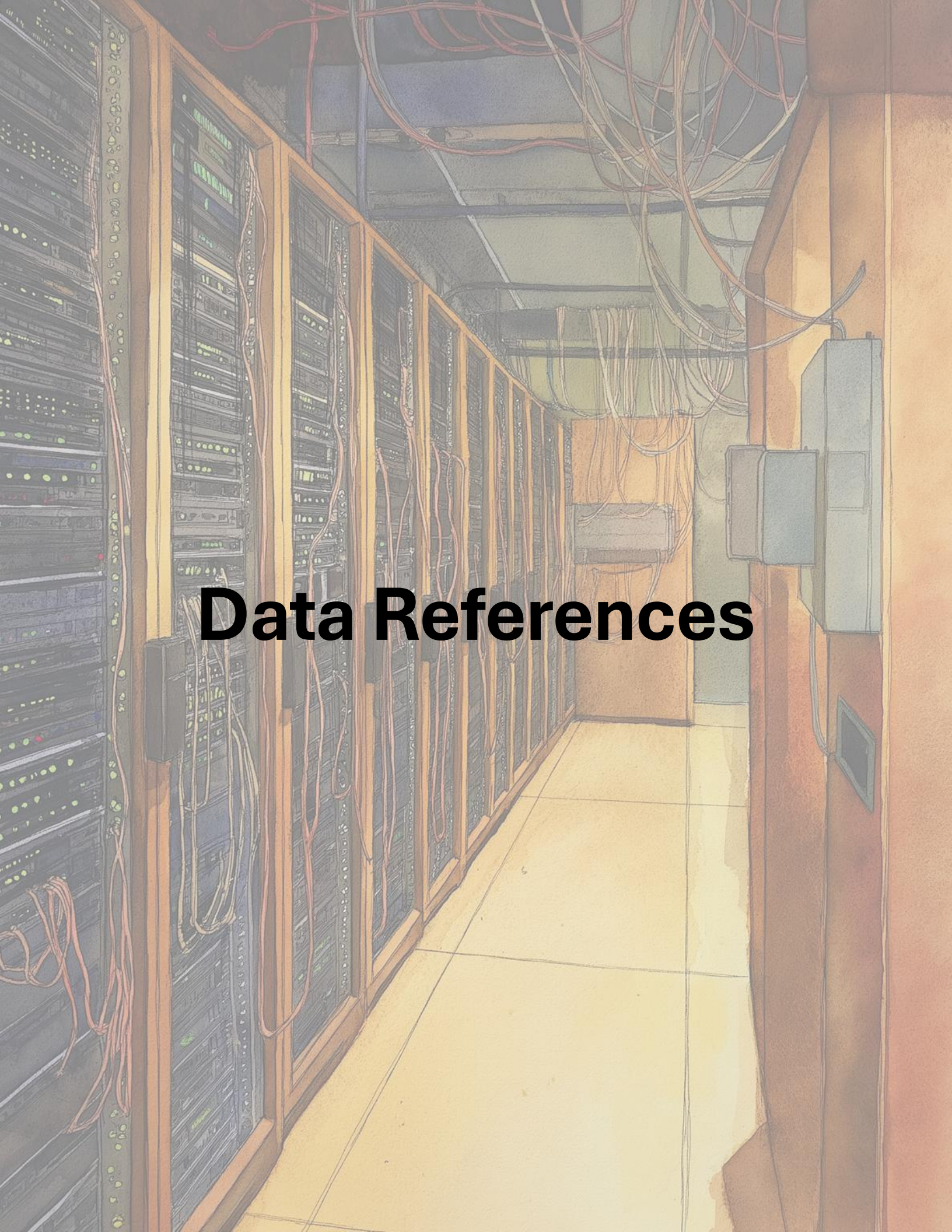
GA	Buford city, Georgia	Gwinnett County, Georgia
GA	Haralson town, Georgia	Coweta County, Georgia
GA	Sharpsburg town, Georgia	Coweta County, Georgia
GA	Turin town, Georgia	Coweta County, Georgia
GA	Senoia city, Georgia	Coweta County, Georgia

The selector module of Abakus calls the names of geographies from the first column of the most recent year of data from the U.S. Census American Community Survey 5-Year Estimates. The user selects the state abbreviation first, which then updates the data validation drop down so that only places in that state are selectable. A crosswalk sheet informs all of this, and details what state and what county a given place is located in. Even if a user wants to focus on a county, they have to first select a place within that county in order to return data on it.

Ideally, as we move Abakus forward the flow of selecting geographies would be smoother, where maybe the user still has to select the state, but then from there, there might be able to select whether they are interested in Places, Counties or County Subdivisions. There could also be additional options to also include data on the County or County Subdivision if the user is primarily interested in places, but also is interested in how data for a place compares to the larger area surrounding them.

The geographies and this crosswalk are pieces of data that I don't think we can automate at this point, and I'm fine producing these sheets annually. The trick will be to build the tool so that these sheets can be changed out annually and the tool sill function seamlessly.

The Selector Module also currently provides users with the ability to select the communities that they want to compare to the community that they are primarily interested in. If possible, I think we should keep this functionality.



Data References

ACS - B19037

Column Name: Estimate

Included Variables	Data Format	Change Between 2010 to 2022?	Nature of Change if Applicable
Total:	Whole Number	None	N/A
Householder under 25 years:	Whole Number	None	N/A
Less than \$10,000	Whole Number	None	N/A
\$10,000 to \$14,999	Whole Number	None	N/A
\$15,000 to \$19,999	Whole Number	None	N/A
\$20,000 to \$24,999	Whole Number	None	N/A
\$25,000 to \$29,999	Whole Number	None	N/A
\$30,000 to \$34,999	Whole Number	None	N/A
\$35,000 to \$39,999	Whole Number	None	N/A
\$40,000 to \$44,999	Whole Number	None	N/A
\$45,000 to \$49,999	Whole Number	None	N/A
\$50,000 to \$59,999	Whole Number	None	N/A
\$60,000 to \$74,999	Whole Number	None	N/A
\$75,000 to \$99,999	Whole Number	None	N/A
\$100,000 to \$124,999	Whole Number	None	N/A
\$125,000 to \$149,999	Whole Number	None	N/A
\$150,000 to \$199,999	Whole Number	None	N/A
\$200,000 or more	Whole Number	None	N/A
Householder 25 to 44 years:	Whole Number	None	N/A
Less than \$10,000	Whole Number	None	N/A
\$10,000 to \$14,999	Whole Number	None	N/A
\$15,000 to \$19,999	Whole Number	None	N/A
\$20,000 to \$24,999	Whole Number	None	N/A
\$25,000 to \$29,999	Whole Number	None	N/A
\$30,000 to \$34,999	Whole Number	None	N/A
\$35,000 to \$39,999	Whole Number	None	N/A
\$40,000 to \$44,999	Whole Number	None	N/A
\$45,000 to \$49,999	Whole Number	None	N/A
\$50,000 to \$59,999	Whole Number	None	N/A
\$60,000 to \$74,999	Whole Number	None	N/A
\$75,000 to \$99,999	Whole Number	None	N/A
\$100,000 to \$124,999	Whole Number	None	N/A
\$125,000 to \$149,999	Whole Number	None	N/A
\$150,000 to \$199,999	Whole Number	None	N/A
\$200,000 or more	Whole Number	None	N/A
Householder 45 to 64 years:	Whole Number	None	N/A
Less than \$10,000	Whole Number	None	N/A
\$10,000 to \$14,999	Whole Number	None	N/A
\$15,000 to \$19,999	Whole Number	None	N/A
\$20,000 to \$24,999	Whole Number	None	N/A
\$25,000 to \$29,999	Whole Number	None	N/A
\$30,000 to \$34,999	Whole Number	None	N/A
\$35,000 to \$39,999	Whole Number	None	N/A
\$40,000 to \$44,999	Whole Number	None	N/A
\$45,000 to \$49,999	Whole Number	None	N/A
\$50,000 to \$59,999	Whole Number	None	N/A
\$60,000 to \$74,999	Whole Number	None	N/A
\$75,000 to \$99,999	Whole Number	None	N/A
\$100,000 to \$124,999	Whole Number	None	N/A
\$125,000 to \$149,999	Whole Number	None	N/A
\$150,000 to \$199,999	Whole Number	None	N/A
\$200,000 or more	Whole Number	None	N/A
Householder 65 years and over:	Whole Number	None	N/A
Less than \$10,000	Whole Number	None	N/A
\$10,000 to \$14,999	Whole Number	None	N/A
\$15,000 to \$19,999	Whole Number	None	N/A
\$20,000 to \$24,999	Whole Number	None	N/A
\$25,000 to \$29,999	Whole Number	None	N/A
\$30,000 to \$34,999	Whole Number	None	N/A
\$35,000 to \$39,999	Whole Number	None	N/A
\$40,000 to \$44,999	Whole Number	None	N/A
\$45,000 to \$49,999	Whole Number	None	N/A
\$50,000 to \$59,999	Whole Number	None	N/A
\$60,000 to \$74,999	Whole Number	None	N/A
\$75,000 to \$99,999	Whole Number	None	N/A
\$100,000 to \$124,999	Whole Number	None	N/A
\$125,000 to \$149,999	Whole Number	None	N/A
\$150,000 to \$199,999	Whole Number	None	N/A
\$200,000 or more	Whole Number	None	N/A

ACS – B25007

Column Name: Estimate

Included Variables	Data Format	Change Between 2010 to 2022?	Nature of Change if Applicable
Total:	Whole Number	None	N/A
Owner occupied:	Whole Number	None	N/A
Householder 15 to 24 years	Whole Number	None	N/A
Householder 25 to 34 years	Whole Number	None	N/A
Householder 35 to 44 years	Whole Number	None	N/A
Householder 45 to 54 years	Whole Number	None	N/A
Householder 55 to 59 years	Whole Number	None	N/A
Householder 60 to 64 years	Whole Number	None	N/A
Householder 65 to 74 years	Whole Number	None	N/A
Householder 75 to 84 years	Whole Number	None	N/A
Householder 85 years and over	Whole Number	None	N/A
Renter occupied:	Whole Number	None	N/A
Householder 15 to 24 years	Whole Number	None	N/A
Householder 25 to 34 years	Whole Number	None	N/A
Householder 35 to 44 years	Whole Number	None	N/A
Householder 45 to 54 years	Whole Number	None	N/A
Householder 55 to 59 years	Whole Number	None	N/A
Householder 60 to 64 years	Whole Number	None	N/A
Householder 65 to 74 years	Whole Number	None	N/A
Householder 75 to 84 years	Whole Number	None	N/A
Householder 85 years and over	Whole Number	None	N/A

ACS – B25118

Column Name: Estimate

Included Variables	Data Format	Change Between 2010 to 2022?	Nature of Change if Applicable
Total:	Whole Number	None	N/A
Owner occupied:	Whole Number	None	N/A
Less than \$5,000	Whole Number	None	N/A
\$5,000 to \$9,999	Whole Number	None	N/A
\$10,000 to \$14,999	Whole Number	None	N/A
\$15,000 to \$19,999	Whole Number	None	N/A
\$20,000 to \$24,999	Whole Number	None	N/A
\$25,000 to \$34,999	Whole Number	None	N/A
\$35,000 to \$49,999	Whole Number	None	N/A
\$50,000 to \$74,999	Whole Number	None	N/A
\$75,000 to \$99,999	Whole Number	None	N/A
\$100,000 to \$149,999	Whole Number	None	N/A
\$150,000 or more	Whole Number	None	N/A
Renter occupied:	Whole Number	None	N/A
Less than \$5,000	Whole Number	None	N/A
\$5,000 to \$9,999	Whole Number	None	N/A
\$10,000 to \$14,999	Whole Number	None	N/A
\$15,000 to \$19,999	Whole Number	None	N/A
\$20,000 to \$24,999	Whole Number	None	N/A
\$25,000 to \$34,999	Whole Number	None	N/A
\$35,000 to \$49,999	Whole Number	None	N/A
\$50,000 to \$74,999	Whole Number	None	N/A
\$75,000 to \$99,999	Whole Number	None	N/A
\$100,000 to \$149,999	Whole Number	None	N/A
\$150,000 or more	Whole Number	None	N/A

ACS – DP04

Column Name: Estimate

Included Variables	Data Format	Change Between 2010 to 2022?	Nature of Change if Applicable
Total housing units	Whole Number	None	N/A
Occupied housing units	Whole Number	None	N/A
Vacant housing units	Whole Number	None	N/A
Homeowner vacancy rate	Whole Number	None	N/A
Rental vacancy rate	Whole Number	None	N/A
1-unit, detached	Whole Number	None	N/A
1-unit, attached	Whole Number	None	N/A
2 units	Whole Number	None	N/A
3 or 4 units	Whole Number	None	N/A
5 to 9 units	Whole Number	None	N/A
10 to 19 units	Whole Number	None	N/A
20 or more units	Whole Number	None	N/A
Mobile home	Whole Number	None	N/A
Boat, RV, van, etc.	Whole Number	None	N/A
Built 2020 or later	Whole Number	None	N/A
Built 2010 to 2019	Whole Number	None	N/A
Built 2000 to 2009	Whole Number	None	N/A
Built 1990 to 1999	Whole Number	None	N/A
Built 1980 to 1989	Whole Number	None	N/A
Built 1970 to 1979	Whole Number	None	N/A
Built 1960 to 1969	Whole Number	None	N/A
Built 1950 to 1959	Whole Number	None	N/A
Built 1940 to 1949	Whole Number	None	N/A
Built 1939 or earlier	Whole Number	None	N/A
Owner-occupied	Whole Number	None	N/A
Renter-occupied	Whole Number	None	N/A
Moved in 2021 or later	Whole Number	None	N/A
Moved in 2018 to 2020	Whole Number	None	N/A
Moved in 2010 to 2017	Whole Number	None	N/A
Moved in 2000 to 2009	Whole Number	None	N/A
Moved in 1990 to 1999	Whole Number	None	N/A
Moved in 1989 and earlier	Whole Number	None	N/A
Less than \$50,000	Whole Number	None	N/A
\$50,000 to \$99,999	Whole Number	None	N/A
\$100,000 to \$149,999	Whole Number	None	N/A
\$150,000 to \$199,999	Whole Number	None	N/A
\$200,000 to \$299,999	Whole Number	None	N/A
\$300,000 to \$499,999	Whole Number	None	N/A
\$500,000 to \$999,999	Whole Number	None	N/A
\$1,000,000 or more	Whole Number	None	N/A
Median (dollars)	Whole Number	None	N/A
Housing units with a mortgage (excluding units where SMOCAPI cannot be computed)	Whole Number	None	N/A
Less than 20.0 percent	Whole Number	None	N/A
20.0 to 24.9 percent	Whole Number	None	N/A
25.0 to 29.9 percent	Whole Number	None	N/A
30.0 to 34.9 percent	Whole Number	None	N/A
35.0 percent or more	Whole Number	None	N/A
Not computed	Whole Number	None	N/A
Occupied units paying rent	Whole Number	None	N/A
Less than \$500	Whole Number	None	N/A
\$500 to \$999	Whole Number	None	N/A
\$1,000 to \$1,499	Whole Number	None	N/A
\$1,500 to \$1,999	Whole Number	None	N/A
\$2,000 to \$2,499	Whole Number	None	N/A
\$2,500 to \$2,999	Whole Number	None	N/A
\$3,000 or more	Whole Number	None	N/A
Median (dollars)	Whole Number	None	N/A
Occupied units paying rent (excluding units where GRAP1 cannot be computed)	Whole Number	None	N/A
Less than 15.0 percent	Whole Number	None	N/A
15.0 to 19.9 percent	Whole Number	None	N/A
20.0 to 24.9 percent	Whole Number	None	N/A
25.0 to 29.9 percent	Whole Number	None	N/A
30.0 to 34.9 percent	Whole Number	None	N/A
35.0 percent or more	Whole Number	None	N/A

ACS – DP05

Column Name: Estimate

Included Variables	Data Format	Change Between 2010 to 2022?	Nature of Change if Applicable
Total population	Whole Number	None	N/A
Under 5 years	Whole Number	None	N/A
5 to 9 years	Whole Number	None	N/A
10 to 14 years	Whole Number	None	N/A
15 to 19 years	Whole Number	None	N/A
20 to 24 years	Whole Number	None	N/A
25 to 34 years	Whole Number	None	N/A
35 to 44 years	Whole Number	None	N/A
45 to 54 years	Whole Number	None	N/A
55 to 59 years	Whole Number	None	N/A
60 to 64 years	Whole Number	None	N/A
65 to 74 years	Whole Number	None	N/A
75 to 84 years	Whole Number	None	N/A
85 years and over	Whole Number	None	N/A
Median age (years)	Whole Number	None	N/A
One race	Whole Number	None	N/A
White	Whole Number	None	N/A
Black or African American	Whole Number	None	N/A
American Indian and Alaska Native	Whole Number	None	N/A
Asian	Whole Number	None	N/A
Native Hawaiian and Other Pacific Islander	Whole Number	None	N/A
Some Other Race	Whole Number	None	N/A
Hispanic or Latino (of any race)	Whole Number	None	N/A
Not Hispanic or Latino	Whole Number	None	N/A

ACS – S1101

Column Name: Total → Estimate

Included Variables	Data Format	Change Between 2010 to 2022?	Nature of Change if Applicable
Total households	Whole Number	None	N/A
Average household size	Whole Number	None	N/A

ACS – S1901

Column Name: Households → Estimate

Included Variables	Data Format	Change Between 2010 to 2022?	Nature of Change if Applicable
Total	Whole Number	None	N/A
Less than \$10,000	Percentage	None	N/A
\$10,000 to \$14,999	Percentage	None	N/A
\$15,000 to \$24,999	Percentage	None	N/A
\$25,000 to \$34,999	Percentage	None	N/A
\$35,000 to \$49,999	Percentage	None	N/A
\$50,000 to \$74,999	Percentage	None	N/A
\$75,000 to \$99,999	Percentage	None	N/A
\$100,000 to \$149,999	Percentage	None	N/A
\$150,000 to \$199,999	Percentage	None	N/A
\$200,000 or more	Percentage	None	N/A
Median income (dollars)	Whole Number	None	N/A

ACS – S2501

Column Name: Occupied Housing Units →
Estimate

Included Variables	Data Format	Change Between 2010 to 2022?	Nature of Change if Applicable
Occupied housing units	Whole Number	None	N/A
1-person household	Whole Number	Format Change	2017 Change From Percentage to Whole Number
2-person household	Whole Number	Format Change	2017 Change From Percentage to Whole Number
3-person household	Whole Number	Format Change	2017 Change From Percentage to Whole Number
4-or-more-person household	Whole Number	Format Change	2017 Change From Percentage to Whole Number
Family households	Whole Number	Format Change	2017 Change From Percentage to Whole Number
Nonfamily households	Whole Number	Format Change	2017 Change From Percentage to Whole Number



Demographics Dashboard

Population

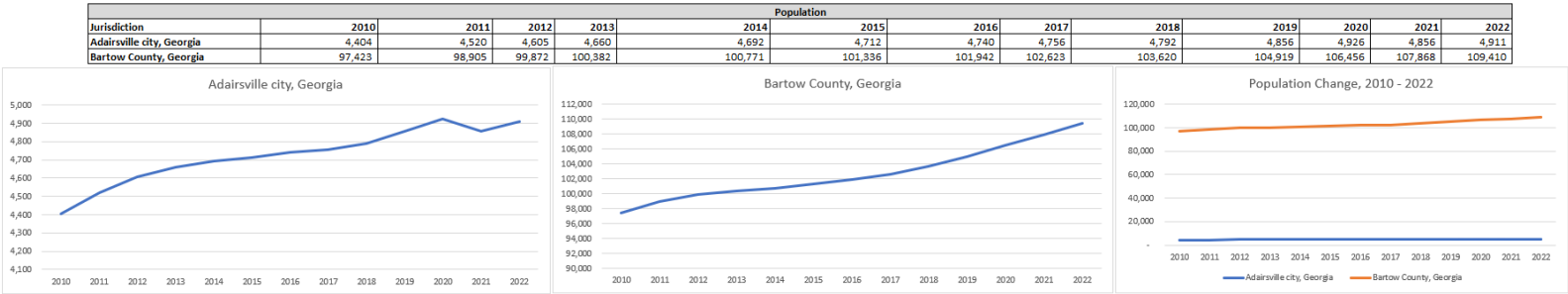


Table:
DP05
Variable(s):
Total population

Transformation:
None, direct report

Two-Period Comparison or Longitudinal:
Longitudinal

Households

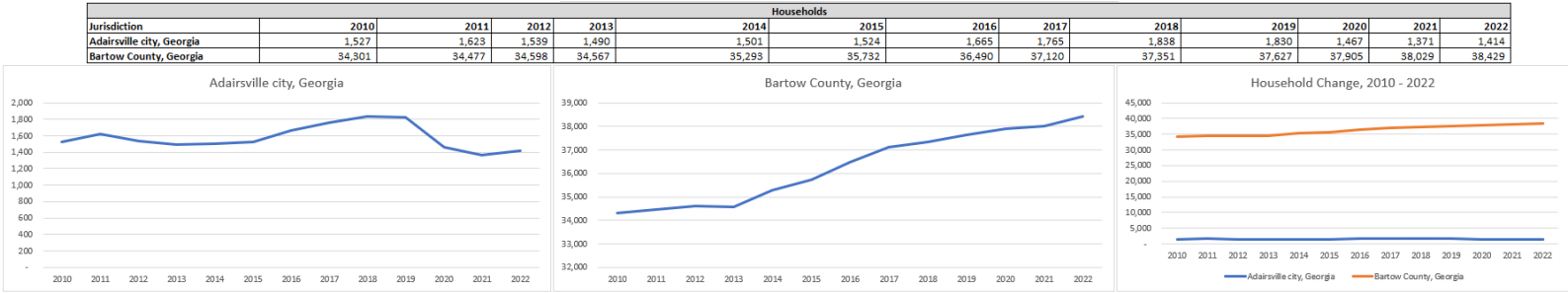


Table:

S1101

Variable(s):

Total households

Transformation:

None, direct report

Two-Period Comparison or Longitudinal:

Longitudinal

Housing Units

Jurisdiction	Housing Units												
	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022
Adairsville city, Georgia	1,826	1,868	1,861	1,854	1,858	1,829	1,915	1,914	1,964	1,967	1,757	1,687	1,771
Bartow County, Georgia	38,884	39,446	39,680	39,715	39,844	39,838	39,935	40,380	40,694	41,072	41,684	41,946	42,731

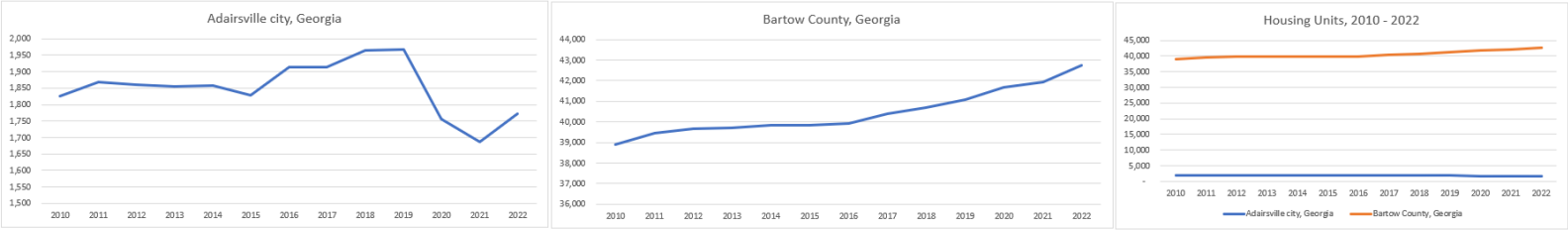


Table:

DP04

Variable(s):

Total housing units

Transformation:

None, direct report

Two-Period Comparison or Longitudinal:

Longitudinal

Housing Unit Occupancy

Housing Unit Occupancy		
Adairsville city, Georgia	2012	2022
Occupied housing units	1,539	1,414
Vacant housing units	322	357

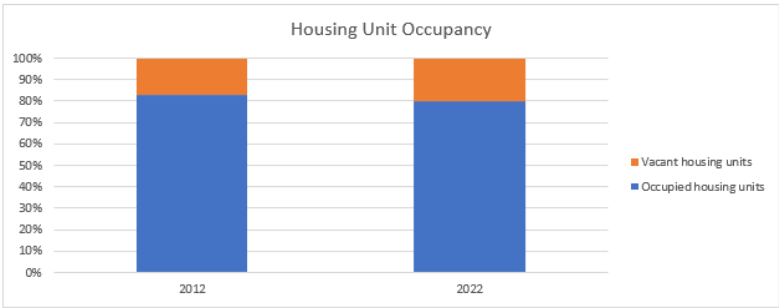


Table:

DP04

Variable(s):

Occupied housing units, Vacant housing units

Transformation:

Both variables divided by Total housing units

Two-Period Comparison or Longitudinal:

Two-Period Comparison

Housing Unit Type

Housing Unit Type		
Adairsville city, Georgia	2012	2022
1-unit, detached	987	1,348
1-unit, attached	198	96
2 units	92	77
3 or 4 units	183	71
5 to 9 units	31	80
10 to 19 units	81	-
20 or more units	15	84
Mobile home	254	15
Boat, RV, van, etc.	20	-

Housing Unit Type (Simplified)		
Adairsville city, Georgia	2012	2022
SFD	987	1,348
SFA	198	96
Small MF	306	228
Large MF	96	84
Other	274	15

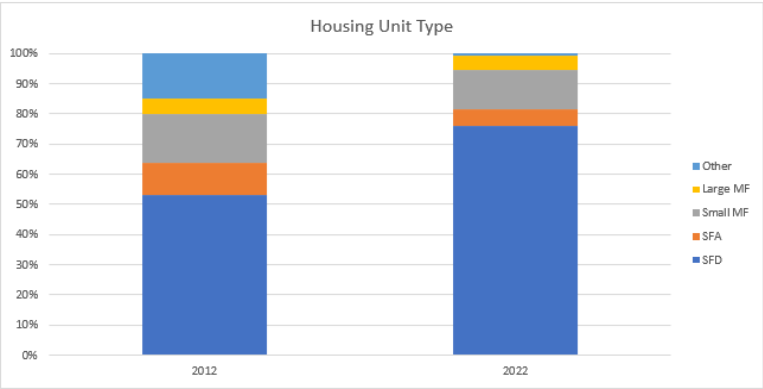


Table:

DP04

Variable(s):

1-unit, detached; 1-unit, attached; 2 units; 3 or 4 units; 5 to 9 units; 10 to 19 units; 20 or more units; Mobile home; Boat, RV, van, etc.

Transformation:

None, direct report, however this could be displayed as percentages too in which case we'd divide all variables by Total housing units.

Additionally, the chart actually displays a simplified version of the data in which 1-unit, detached becomes SFD, 1-unit, attached becomes SFA, 2 units through 10 to 19 units becomes Small MF, 20 or more units becomes Large MF, and the last two become Other.

Two-Period Comparison or Longitudinal:

Two-Period Comparison

Year Built

Year Built (2022 Data)		
	Adairsville city, Georgia	Bartow County, Georgia
Built 1939 or earlier	1%	3%
Built 1940 to 1949	6%	3%
Built 1950 to 1959	4%	5%
Built 1960 to 1969	9%	7%
Built 1970 to 1979	5%	9%
Built 1980 to 1989	10%	17%
Built 1990 to 1999	15%	23%
Built 2000 to 2009	37%	22%
Built 2010 to 2019	7%	9%
Built 2020 or later	6%	2%

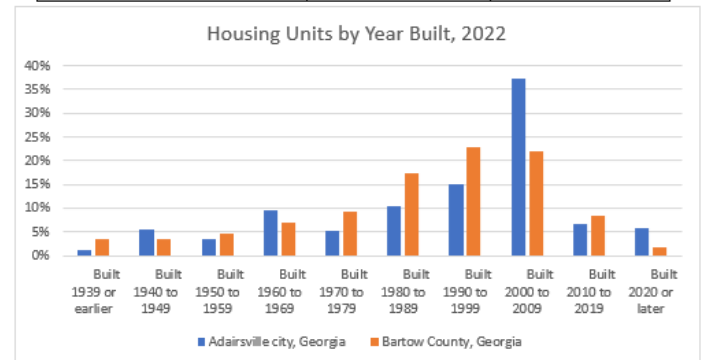


Table:

DP04

Variable(s):

Built 1939 or earlier, Built 1940 to 1949, Built 1950 to 1959, Built 1960 to 1969, Built 1970 to 1979, Built 1980 to 1989, Built 1990 to 1999, Built 2000 to 2009, Built 2010 to 2019, Built 2020 or later*

Transformation:

All variables divided by Total housing units

Two-Period Comparison or Longitudinal:

Neither as this data should only come from the most recent year. Census fucks this up every year so it makes it impossible to look at this data across time because things stop making sense. This is one of two pieces of data that should only be returned for the most recent year no matter what years the user is selecting.

Additional Note:

Over time the Census will add new bins so this table we're generating will need to be updated as Census rolls out new bins for more recent years

Year Moved In

Year Moved into Current Housing Unit (2022 Data)		
	Adairsville city, Georgia	Bartow County, Georgia
Moved in 1989 and earlier	0%	8%
Moved in 1990 to 1999	10%	13%
Moved in 2000 to 2009	21%	19%
Moved in 2010 to 2017	44%	35%
Moved in 2018 to 2020	18%	20%
Moved in 2021 or later	7%	4%

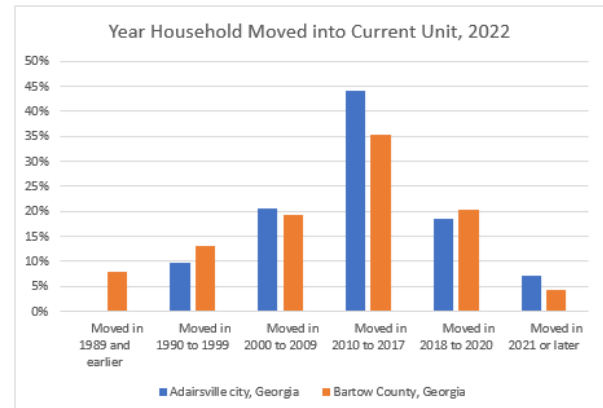


Table:

DP04

Variable(s):

Moved in 1989 and earlier, Moved in 1990 to 1999, Moved in 2000 to 2009, Moved in 2010 to 2017, Moved in 2018 to 2020, Moved in 2021 or later*

Transformation:

All variables divided by Occupied housing units

Two-Period Comparison or Longitudinal:

Neither as this data should only come from the most recent year. Census fucks this up every year so it makes it impossible to look at this data across time because things stop making sense. This is one of two pieces of data that should only be returned for the most recent year no matter what years the user is selecting.

Additional Note:

Over time the Census will add new bins so this table we're generating will need to be updated as Census rolls out new bins for more recent years

Tenure

Tenure		
Adairsville city, Georgia	2012	2022
Owner-occupied	62%	72%
Renter-occupied	38%	28%

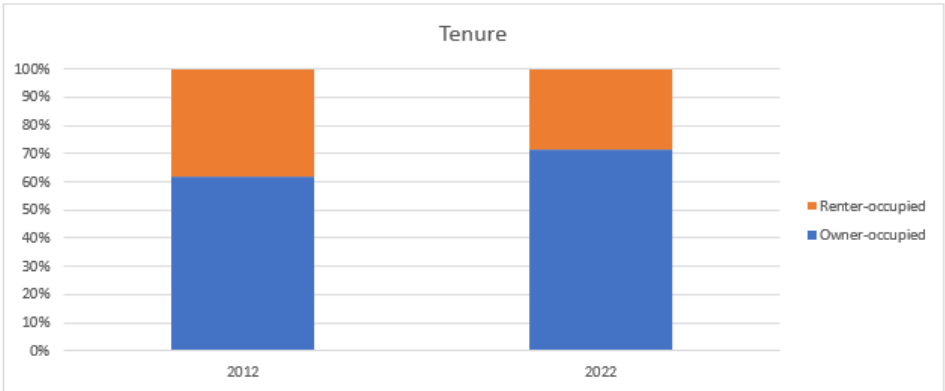


Table:

DP04

Variable(s):

Owner-occupied, Renter-occupied

Transformation:

Both variables divided by Occupied housing units

Two-Period Comparison or Longitudinal:

Two-Period Comparison

Median Home Value

Median Home Value		
Jurisdiction	2012	2022
Adairsville city, Georgia	\$123,300	\$186,900
Bartow County, Georgia	\$137,600	\$237,000

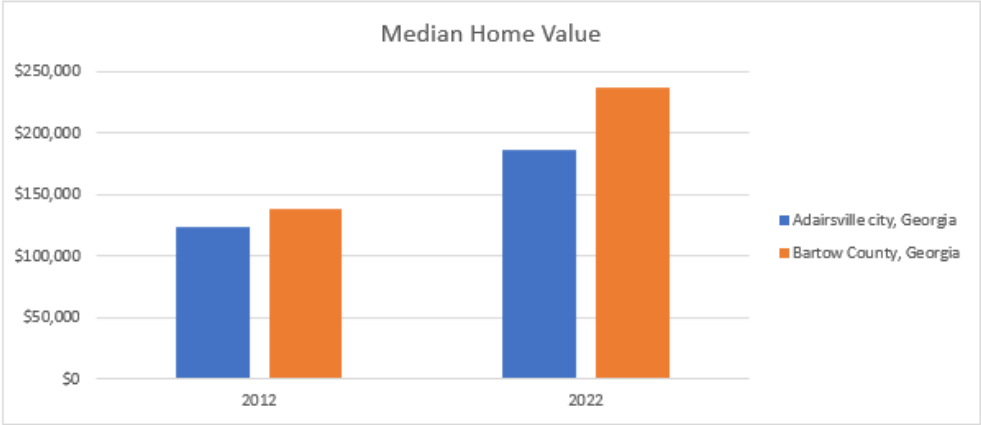


Table:

DP04

Variable(s):

Median (dollars)*

Transformation:

None, direct report

Two-Period Comparison or Longitudinal:

Two-Period Comparison

Additional Note:

Median (dollars) is a variable that is essentially repeated twice in the dataset. The first time it appears it refers to median home value, the second time it appears it refers to median rent paid.

Home Value Distribution

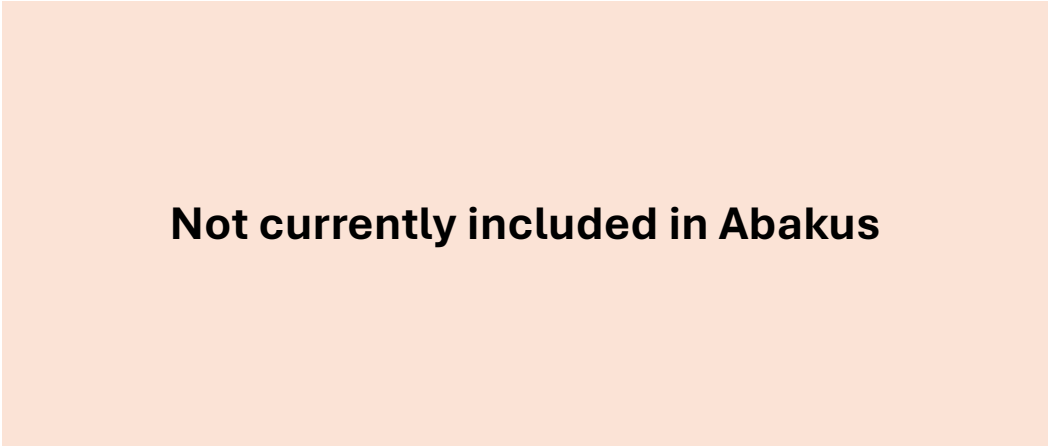


Table:

DP04

Variable(s):

Less than \$50,000; \$50,000 to \$99,999; \$100,000 to \$149,999; \$150,000 to \$199,999; \$200,000 to \$299,999; \$300,000 to \$499,999; \$500,000 to \$999,999; \$1,000,000 or more

Transformation:

None, direct report OR as a percentage by dividing all variables by Owner-occupied

Two-Period Comparison or Longitudinal:

Two-Period Comparison

Median Rent

Median Rent		
Jurisdiction	2012	2022
Adairsville city, Georgia	\$697	\$1,190
Bartow County, Georgia	\$774	\$1,090

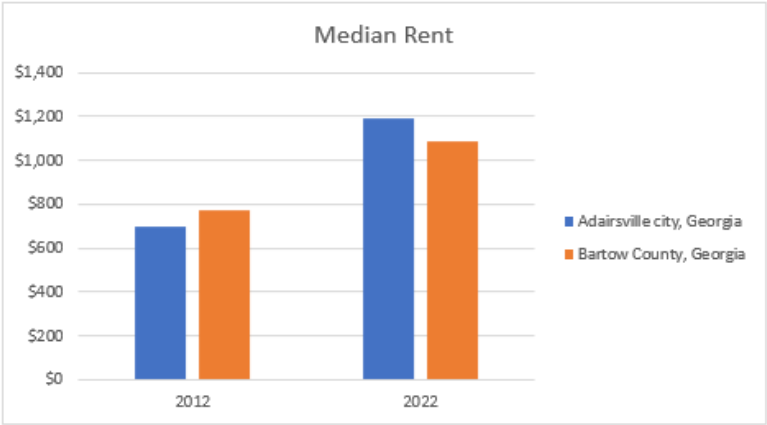


Table:

DP04

Variable(s):

Median (dollars)*

Transformation:

None, direct report

Two-Period Comparison or Longitudinal:

Two-Period Comparison

Additional Note:

Median (dollars) is a variable that is essentially repeated twice in the dataset. The first time it appears it refers to median home value, the second time it appears it refers to median rent paid.

Rent Paid Distribution

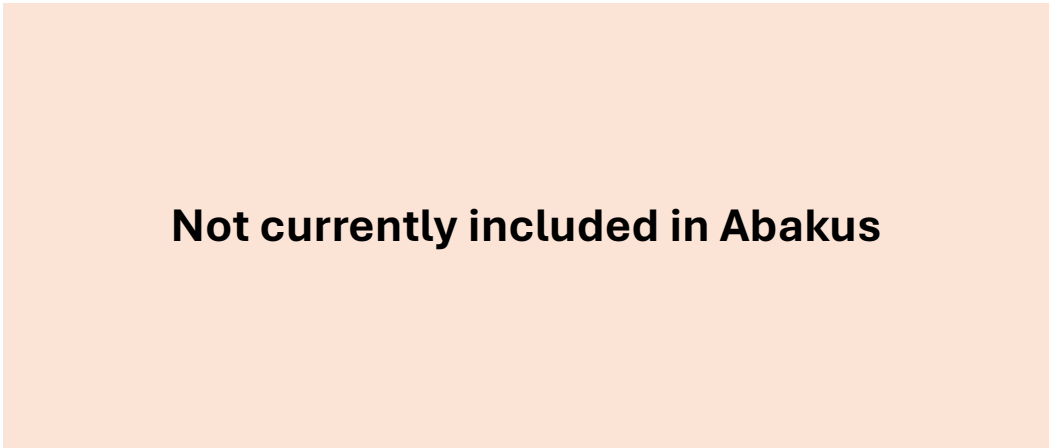


Table:

DP04

Variable(s):

Less than \$500; \$500 to \$999; \$1,000 to \$1,499; \$1,500 to \$1,999; \$2,000 to \$2,499; \$2,500 to \$2,999; \$3,000 or more

Transformation:

None, direct report OR as a percentage by dividing all variables by Renter-occupied

Two-Period Comparison or Longitudinal:

Two-Period Comparison

Owner Cost Burden

Owner Cost Burden		
Adairsville city, Georgia	2012	2022
Less than 20.0 percent	31%	75%
20.0 to 24.9 percent	32%	6%
25.0 to 29.9 percent	14%	8%
30.0 to 34.9 percent	2%	1%
35.0 percent or more	22%	9%

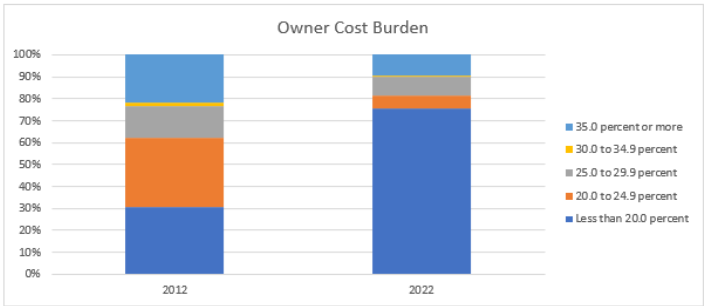


Table:

DP04

Variable(s):

Housing units with a mortgage (excluding units where SMOCAPI cannot be computed), Less than 20.0 percent, 20.0 to 24.9 percent, 25.0 to 29.9 percent, 30.0 to 34.9 percent, 35.0 percent or more

Transformation:

Divide the last five variables by the first variable, Housing units with a mortgage (excluding units where SMOCAPI cannot be computed)

Two-Period Comparison or Longitudinal:

Two-Period Comparison

Renter Cost Burden

Renter Cost Burden		
Adairsville city, Georgia	2012	2022
Less than 15.0 percent	4%	13%
15.0 to 19.9 percent	12%	8%
20.0 to 24.9 percent	6%	30%
25.0 to 29.9 percent	10%	17%
30.0 to 34.9 percent	12%	0%
35.0 percent or more	56%	33%

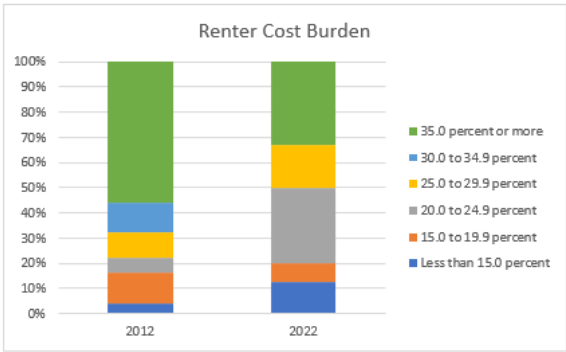


Table:

DP04

Variable(s):

Occupied units paying rent (excluding units where GRAPI cannot be computed), Less than 15.0 percent, 15.0 to 19.9 percent, 20.0 to 24.9 percent, 25.0 to 29.9 percent, 30.0 to 34.9 percent, 35.0 percent or more

Transformation:

Divide the last six variables by the first variable, Housing units paying rent (excluding units where SMOCAPI cannot be computed)

Two-Period Comparison or Longitudinal:

Two-Period Comparison

Age by Cohort

Age Groups		
Adairsville city, Georgia	2012	2022
Under 5 years	7%	8%
5 to 9 years	14%	7%
10 to 14 years	10%	8%
15 to 19 years	7%	8%
20 to 24 years	4%	3%
25 to 34 years	18%	6%
35 to 44 years	17%	18%
45 to 54 years	10%	11%
55 to 59 years	4%	2%
60 to 64 years	4%	5%
65 to 74 years	3%	23%
75 to 84 years	2%	2%
85 years and over	0%	0%

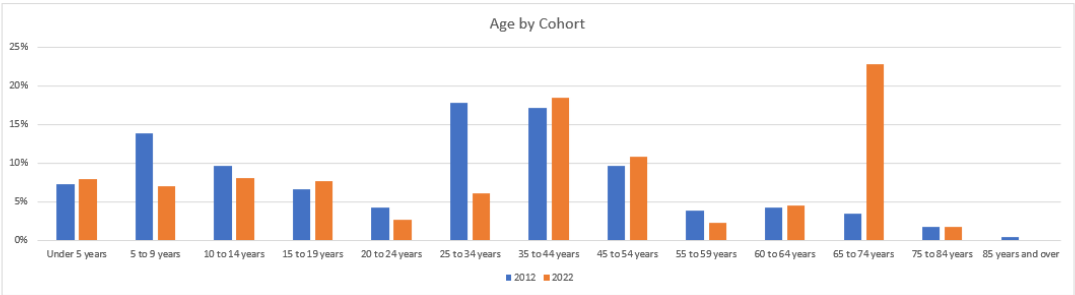


Table:

DP05

Variable(s):

Under 5 years, 5 to 9 years, 10 to 14 years, 15 to 19 years, 20 to 24 years, 25 to 34 years, 35 to 44 years, 45 to 54 years, 55 to 59 years, 60 to 64 years, 65 to 74 years, 75 to 84 years, 85 years and over

Transformation:

Divide all variables by Total population

Two-Period Comparison or Longitudinal:

Two-Period Comparison

Age by Cohort Simplified

Age Group Simplified		
Adairsville city, Georgia	2012	2022
Under 20	37%	31%
20 to 45	39%	27%
45 to 65	18%	18%
65 and Over	6%	25%

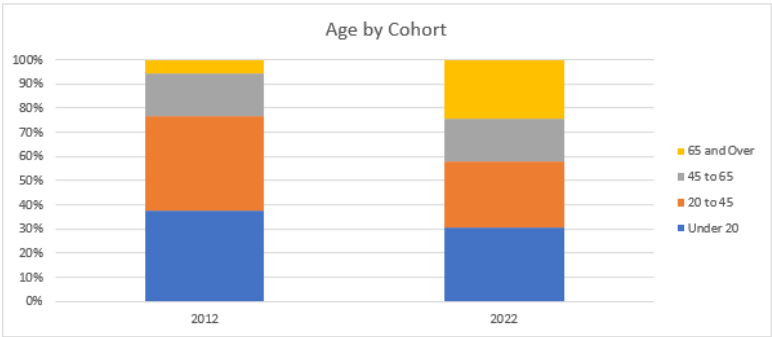


Table:

DP05

Variable(s):

Under 5 years, 5 to 9 years, 10 to 14 years, 15 to 19 years, 20 to 24 years, 25 to 34 years, 35 to 44 years, 45 to 54 years, 55 to 59 years, 60 to 64 years, 65 to 74 years, 75 to 84 years, 85 years and over

Transformation:

Add the percentages to create new groups called Under 20, 20 to 45, 45 to 65 and 65 and Over

Two-Period Comparison or Longitudinal:

Two-Period Comparison

Median Age

Median Age		
Jurisdiction	2012	2022
Adairsville city, Georgia	30.6	40.1
Bartow County, Georgia	36.7	38.2

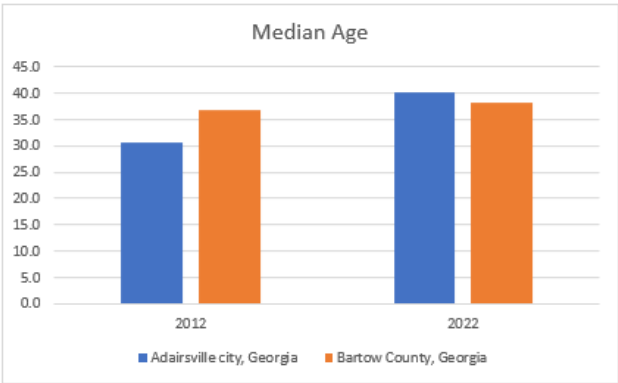


Table:

DP05

Variable(s):

Median age (years)

Transformation:

None, direct report

Two-Period Comparison or Longitudinal:

Two-Period Comparison

Race

Race		
Adairsville city, Georgia	2012	2022
White	74.0%	73.9%
Black or African American	17.1%	5.6%
American Indian and Alaska Native	1.1%	0.0%
Asian	2.1%	0.0%

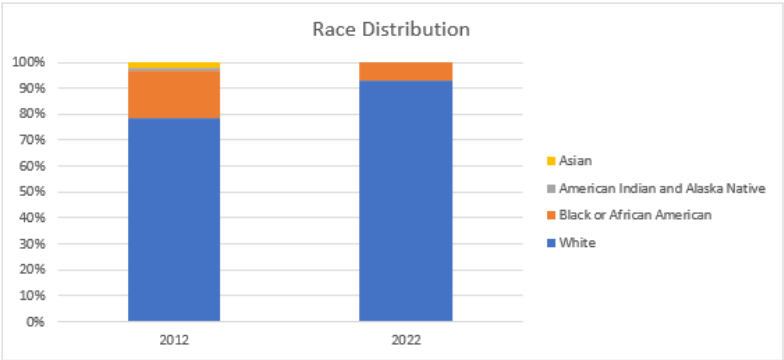


Table:

DP05

Variable(s):

One race, White, Black or African American, American Indian and Alaska Native, Asian, Native Hawaiian and Other Pacific Islander, Some Other Race

Transformation:

Divide each of the six racial categories by One race

Two-Period Comparison or Longitudinal:

Two-Period Comparison

Hispanic Ethnicity

Hispanic Ethnicity		
Adairsville city, Georgia	2012	2022
Hispanic or Latino (of any race)	6.5%	6.6%
Not Hispanic or Latino	93.5%	93.4%

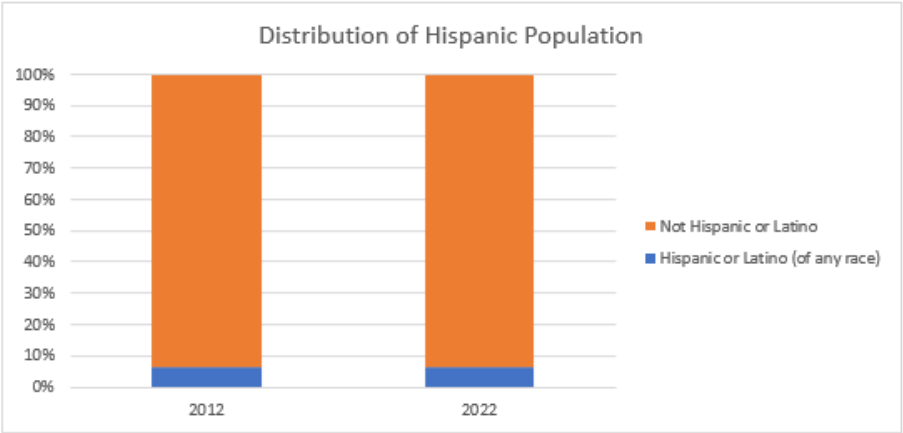


Table:

DP05

Variable(s):

Hispanic or Latino (of any race), Not Hispanic or Latino

Transformation:

Divide both variables by Total population

Two-Period Comparison or Longitudinal:

Two-Period Comparison

Household Income

Household Income		
Adairsville city, Georgia	2012	2022
Less than \$10,000	9%	7%
\$10,000 to \$14,999	9%	0%
\$15,000 to \$24,999	12%	4%
\$25,000 to \$34,999	13%	0%
\$35,000 to \$49,999	23%	9%
\$50,000 to \$74,999	17%	29%
\$75,000 to \$99,999	10%	18%
\$100,000 to \$149,999	5%	31%
\$150,000 to \$199,999	1%	2%
\$200,000 or more	1%	0%

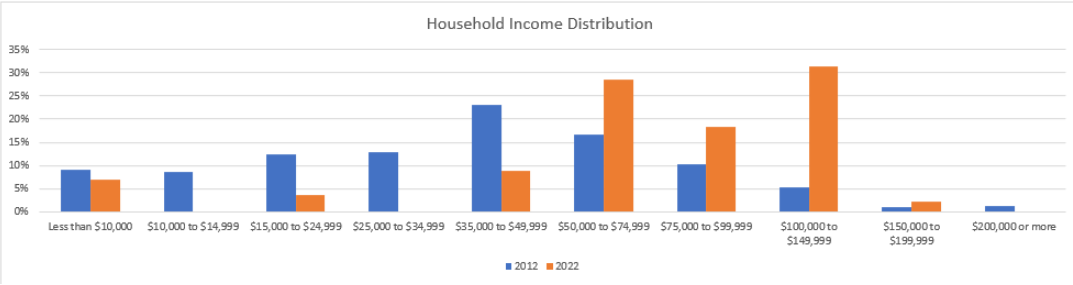


Table:

S1901

Variable(s):

Total, Less than \$10,000, \$10,000 to \$14,999, \$15,000 to \$24,999, \$25,000 to \$34,999, \$35,000 to \$49,999, \$50,000 to \$74,999, \$75,000 to \$99,999, \$100,000 to \$149,999, \$150,000 to \$199,999, \$200,000 or more

Transformation:

Divide all income descriptive variables by Total

Two-Period Comparison or Longitudinal:

Two-Period Comparison

Household Income Simplified

Household Income (Simplified)		
Adairsville city, Georgia	2012	2022
Less than \$35,000	43%	11%
\$35,000 to \$49,000	23%	9%
\$50,000 to \$74,999	17%	29%
\$75,000 to \$99,999	10%	18%
\$100,000 to \$149,999	5%	31%
\$150,000 or more	2%	2%

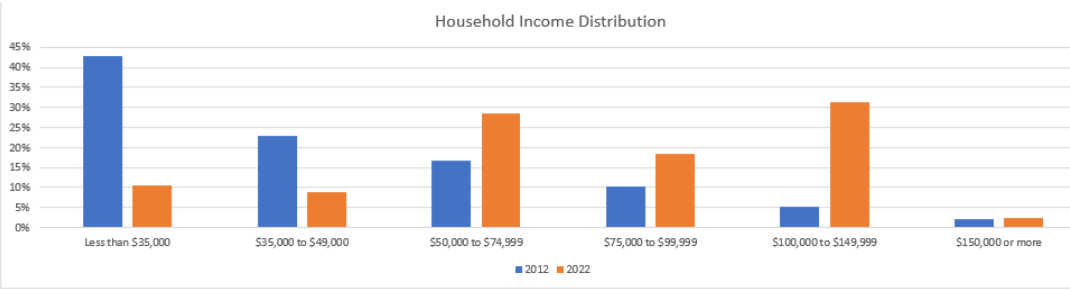


Table:

S1901

Variable(s):

Total, Less than \$10,000, \$10,000 to \$14,999, \$15,000 to \$24,999, \$25,000 to \$34,999, \$35,000 to \$49,999, \$50,000 to \$74,999, \$75,000 to \$99,999, \$100,000 to \$149,999, \$150,000 to \$199,999, \$200,000 or more

Transformation:

Add the percentages to create six new bins: Less than \$35,000, \$35,000 to \$50,000, \$50,000 to \$75,000, \$75,000 to \$100,000, \$100,000 to \$150,000, and \$150,000 or more

Two-Period Comparison or Longitudinal:

Two-Period Comparison

Median Household Income

Median Household Income		
Jurisdiction	2012	2022
Adairsville city, Georgia	\$40,565	\$90,062
Bartow County, Georgia	\$49,543	\$74,812

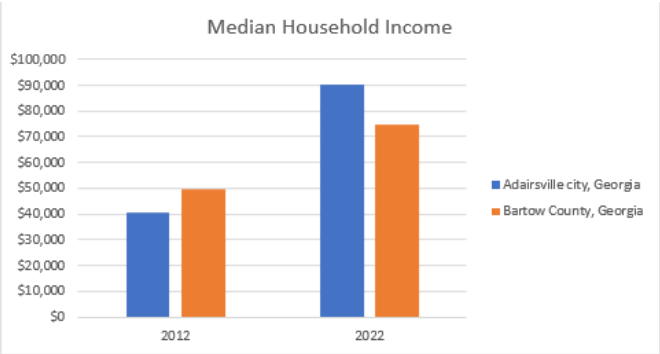


Table:

S1901

Variable(s):

Median income (dollars)

Transformation:

None, direct report

Two-Period Comparison or Longitudinal:

Two-Period Comparison

Tenure by Age

Tenure by Age				
Adairsville city, Georgia				
Age Cohort	2012		2022	
	Owner	Renter	Owner	Renter
Householder 15 to 24 years	1%	4%	0%	11%
Householder 25 to 34 years	17%	30%	14%	13%
Householder 35 to 44 years	26%	43%	26%	55%
Householder 45 to 54 years	20%	5%	33%	0%
Householder 55 to 59 years	12%	3%	3%	0%
Householder 60 to 64 years	12%	7%	11%	16%
Householder 65 to 74 years	8%	3%	11%	4%
Householder 75 to 84 years	3%	3%	2%	0%
Householder 85 years and over	0%	3%	0%	0%

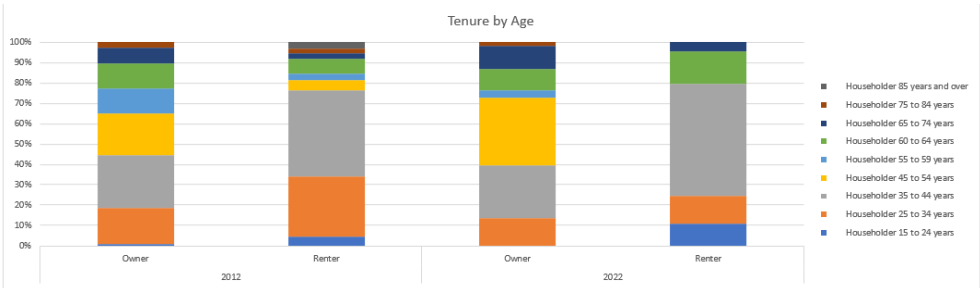


Table:

B25007

Variable(s):

Owner occupied:, Householder 15 to 24 years, Householder 25 to 34 years, Householder 35 to 44 years, Householder 45 to 54 years, Householder 55 to 59 years, Householder 60 to 64 years, Householder 65 to 74 years, Householder 75 to 84 years, Householder 85 years and over

Renter occupied:, Householder 15 to 24 years, Householder 25 to 34 years, Householder 35 to 44 years, Householder 45 to 54 years, Householder 55 to 59 years, Householder 60 to 64 years, Householder 65 to 74 years, Householder 75 to 84 years, Householder 85 years and over

Transformation:

Divide the householder age groups by either Owner occupied: or Renter occupied: depending on what section of the table they fall under

Two-Period Comparison or Longitudinal:

Two-Period Comparison

Tenure by Income

Tenure by Income					
Adairsville city, Georgia					
Income Group	2012		2022		
	Owner	Renter	Owner	Renter	
Less than \$5,000		2%	8%	5%	11%
\$5,000 to \$9,999		0%	13%	0%	0%
\$10,000 to \$14,999		5%	15%	0%	0%
\$15,000 to \$19,999		7%	14%	1%	10%
\$20,000 to \$24,999		2%	5%	0%	0%
\$25,000 to \$34,999		7%	23%	0%	0%
\$35,000 to \$49,999		31%	10%	8%	11%
\$50,000 to \$74,999		22%	8%	25%	37%
\$75,000 to \$99,999		14%	4%	18%	20%
\$100,000 to \$149,999		9%	0%	39%	11%
\$150,000 or more		3%	0%	3%	0%

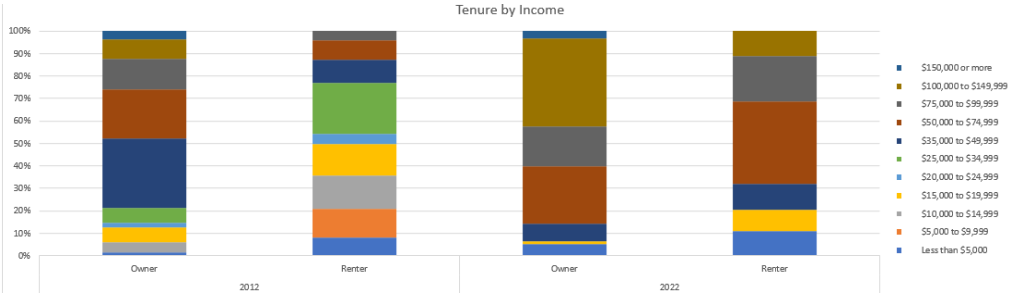


Table:

B25118

Variable(s):

Owner occupied:, Less than \$5,000, \$5,000 to \$9,999, \$10,000 to \$14,999, \$15,000 to \$19,999, \$20,000 to \$24,999, \$25,000 to \$34,999, \$35,000 to \$49,999, \$50,000 to \$74,999, \$75,000 to \$99,999, \$100,000 to \$149,999, \$150,000 or more

Renter occupied:, Less than \$5,000, \$5,000 to \$9,999, \$10,000 to \$14,999, \$15,000 to \$19,999, \$20,000 to \$24,999, \$25,000 to \$34,999, \$35,000 to \$49,999, \$50,000 to \$74,999, \$75,000 to \$99,999, \$100,000 to \$149,999, \$150,000 or more

Transformation:

Divide the householder income groups by either Owner occupied: or Renter occupied: depending on what section of the table they fall under

Two-Period Comparison or Longitudinal:

Two-Period Comparison

Household Size

Household Size		
Adairsville city, Georgia	2012	2022
1-person household	25%	13%
2-person household	29%	35%
3-person household	19%	17%
4-or-more-person household	27%	36%

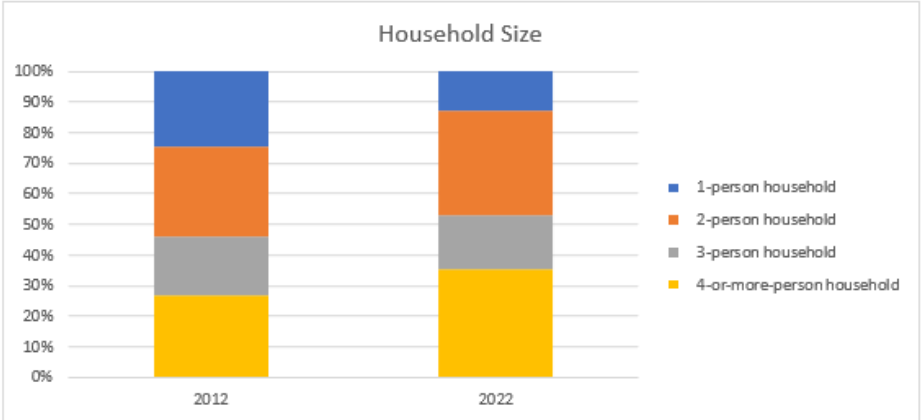


Table:

S2501

Variable(s):

Occupied housing units, 1-person household, 2-person household, 3-person household, 4-or-more-person household

Transformation:

For 2017-Present: Divide the household size figures by Occupied housing units

For 2016 and Earlier: Multiply the household size percentages by Occupied housing units

Two-Period Comparison or Longitudinal:

Two-Period Comparison

Household Type

Household Type		
Adairsville city, Georgia	2012	2022
Family households	72%	77%
Nonfamily households	28%	23%

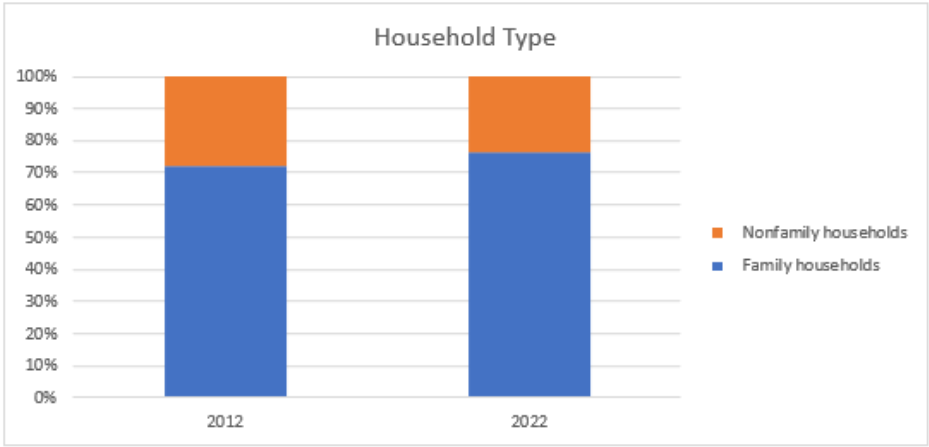


Table:

S2501

Variable(s):

Occupied housing units, Family households, Nonfamily households

Transformation:

For 2017-Present: Divide the household size figures by Occupied housing units

For 2016 and Earlier: Multiply the household size percentages by Occupied housing units

Two-Period Comparison or Longitudinal:

Two-Period Comparison

An aerial, watercolor-style illustration of a scenic lakeside community. A winding river or lake flows through the center, flanked by dense green forests. Numerous houses of various architectural styles are nestled among the trees, some with large porches and gabled roofs. Winding paths lead through the landscape. In the background, rolling hills and mountains are visible under a soft, cloudy sky. The overall tone is peaceful and idyllic.

Comparative Communities

Population

	Population												
Jurisdiction	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022
Adairsville city, Georgia	4,404	4,520	4,605	4,660	4,692	4,712	4,740	4,756	4,792	4,856	4,926	4,856	4,911
Buford city, Georgia	12,032	12,335	12,575	12,754	12,864	13,244	13,927	14,662	15,185	16,143	16,003	16,811	17,222
Haralson town, Georgia	157	168	179	191	197	126	149	120	165	185	203	206	236
Sharpsburg town, Georgia	387	365	398	407	459	538	574	410	402	461	483	519	380
Turin town, Georgia	331	384	339	372	427	412	400	332	571	367	377	342	378
Senoia city, Georgia	3,062	3,174	3,343	3,479	3,627	3,773	3,889	4,027	4,160	4,256	4,386	4,862	5,115

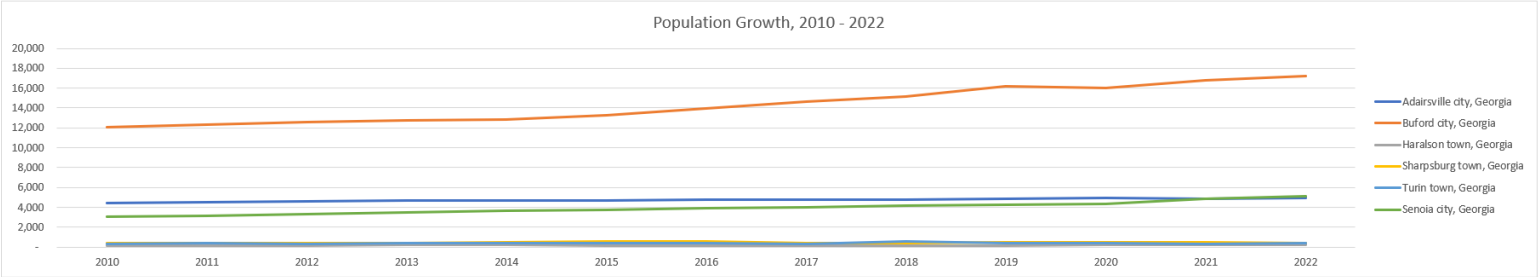


Table:

DP05

Variable(s):

Total population

Transformation:

None, direct report

Two-Period Comparison or Longitudinal:

Longitudinal

Households

Households													
Jurisdiction	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022
Adairsville city, Georgia	1,527	1,623	1,539	1,490	1,501	1,524	1,665	1,765	1,838	1,830	1,467	1,371	1,414
Buford city, Georgia	4,016	4,035	3,918	3,999	4,065	4,275	4,349	4,708	4,818	5,003	4,727	4,998	5,019
Haralson town, Georgia	63	69	61	60	64	57	63	57	70	79	84	86	101
Sharpsburg town, Georgia	149	143	149	153	162	187	210	169	166	199	189	220	163
Turin town, Georgia	113	127	117	128	148	141	138	114	166	121	124	114	134
Senoia city, Georgia	1,159	1,218	1,320	1,330	1,308	1,373	1,323	1,344	1,367	1,387	1,479	1,706	1,796

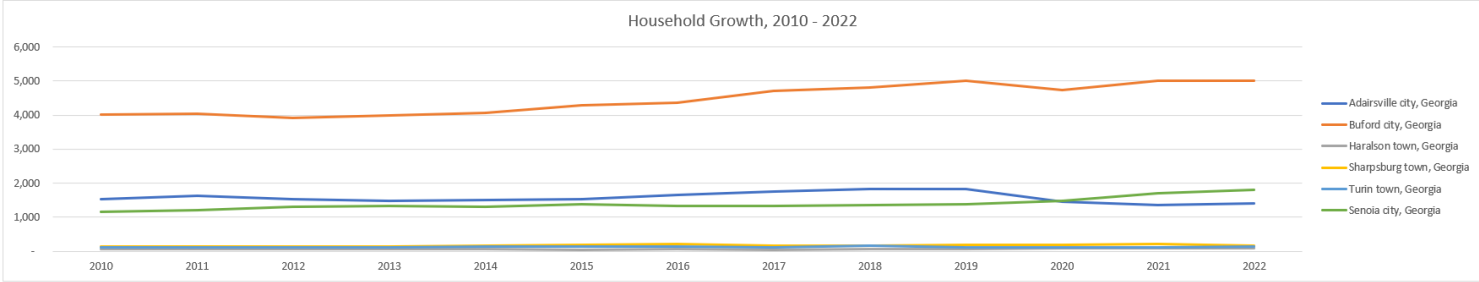


Table:

S1101

Variable(s):

Total households

Transformation:

None, direct report

Two-Period Comparison or Longitudinal:

Longitudinal

Employment

County Employment													
Jurisdiction	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022
Bartow County, Georgia	No Data	31,410	31,357	31,393	32,602	34,328	35,317	36,722	38,736	40,880	39,740	41,213	42,582
Gwinnett County, Georgia	No Data	301,896	308,417	311,894	325,273	334,508	342,828	352,117	354,014	361,947	344,862	358,314	373,486
Coweta County, Georgia	No Data	30,250	30,906	32,735	35,175	36,761	37,140	37,789	39,708	40,544	39,181	42,297	45,956
Coweta County, Georgia	No Data	30,250	30,906	32,735	35,175	36,761	37,140	37,789	39,708	40,544	39,181	42,297	45,956
Coweta County, Georgia	No Data	30,250	30,906	32,735	35,175	36,761	37,140	37,789	39,708	40,544	39,181	42,297	45,956

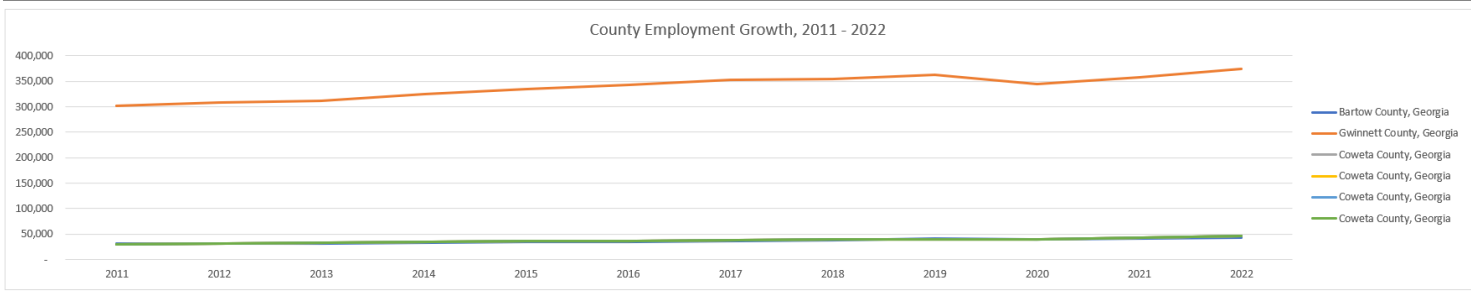


Table:

BLS Employment Data

Variable(s):

annual_avg_emplvl

Transformation:

None, direct report

Two-Period Comparison or Longitudinal:

Longitudinal

Housing Units

Housing Units													
Jurisdiction	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022
Adairsville city, Georgia	1,826	1,868	1,861	1,854	1,858	1,829	1,915	1,914	1,964	1,967	1,757	1,687	1,771
Buford city, Georgia	4,772	4,808	4,709	4,862	4,858	4,929	4,877	5,192	5,199	5,431	5,358	5,578	5,800
Haralson town, Georgia	72	79	68	68	72	65	67	62	73	82	87	88	105
Sharpsburg town, Georgia	161	157	161	173	175	206	229	187	174	207	195	220	166
Turin town, Georgia	116	132	119	136	156	148	143	118	167	125	129	125	144
Senolia city, Georgia	1,257	1,286	1,476	1,474	1,454	1,503	1,468	1,417	1,439	1,440	1,593	1,785	1,892

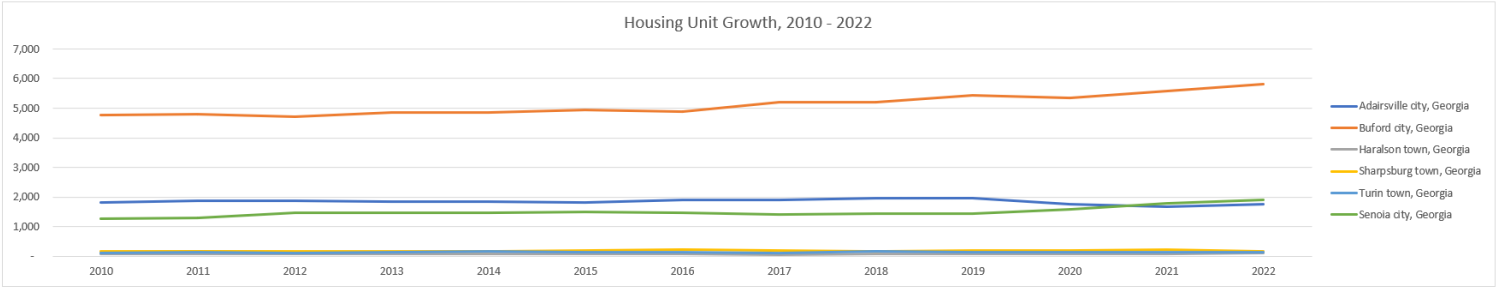


Table:

DP04

Variable(s):

Total housing units

Transformation:

None, direct report

Two-Period Comparison or Longitudinal:

Longitudinal

Housing Unit Occupancy

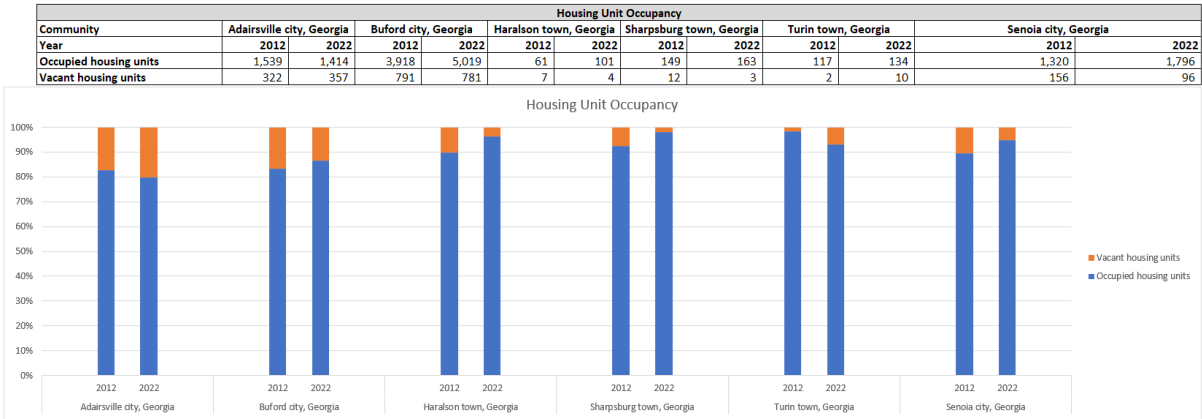


Table:

DP04

Variable(s):

Occupied housing units, Vacant housing units

Transformation:

Both variables divided by Total housing units

Two-Period Comparison or Longitudinal:

Two-Period Comparison

Housing Unit Type

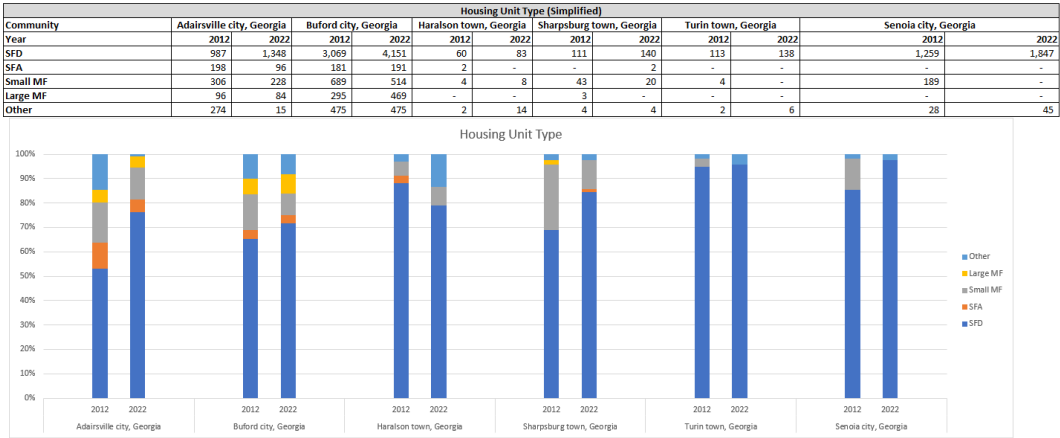


Table:

DP04

Variable(s):

1-unit, detached; 1-unit, attached; 2 units; 3 or 4 units; 5 to 9 units; 10 to 19 units; 20 or more units; Mobile home; Boat, RV, van, etc.

Transformation:

None, direct report, however this could be displayed as percentages too in which case we'd divide all variables by Total housing units.

Additionally, the chart actually displays a simplified version of the data in which 1-unit, detached becomes SFD, 1-unit, attached becomes SFA, 2 units through 10 to 19 units becomes Small MF, 20 or more units becomes Large MF, and the last two become Other.

Two-Period Comparison or Longitudinal:

Two-Period Comparison

Year Built

Year Built (2022 Data)						
	Adairsville city, Georgia	Buford city, Georgia	Haralson town, Georgia	Sharpsburg town, Georgia	Turin town, Georgia	Senoia city, Georgia
Built 1939 or earlier	1%	4%	43%	23%	13%	6%
Built 1940 to 1949	6%	2%	0%	0%	0%	0%
Built 1950 to 1959	4%	8%	0%	2%	3%	0%
Built 1960 to 1969	9%	7%	0%	9%	5%	3%
Built 1970 to 1979	5%	10%	7%	8%	11%	4%
Built 1980 to 1989	10%	16%	21%	19%	2%	9%
Built 1990 to 1999	15%	15%	20%	30%	3%	12%
Built 2000 to 2009	37%	17%	6%	8%	27%	40%
Built 2010 to 2019	7%	18%	4%	1%	22%	24%
Built 2020 or later	6%	2%	0%	0%	13%	1%

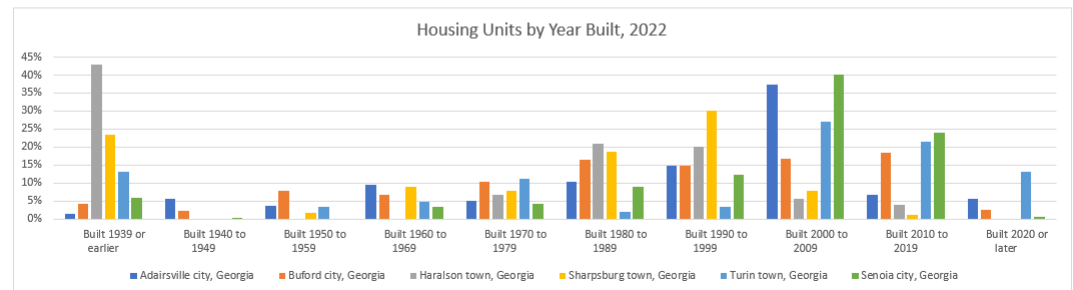


Table:

DP04

Variable(s):

Built 1939 or earlier, Built 1940 to 1949, Built 1950 to 1959, Built 1960 to 1969, Built 1970 to 1979, Built 1980 to 1989, Built 1990 to 1999, Built 2000 to 2009, Built 2010 to 2019, Built 2020 or later*

Transformation:

All variables divided by Total housing units

Two-Period Comparison or Longitudinal:

Neither as this data should only come from the most recent year. Census fucks this up every year so it makes it impossible to look at this data across time because things stop making sense. This is one of two pieces of data that should only be returned for the most recent year no matter what years the user is selecting.

Additional Note:

Over time the Census will add new bins so this table we're generating will need to be updated as Census rolls out new bins for more recent years

Year Moved In

	Adairsville city, Georgia	Buford city, Georgia	Haralson town, Georgia	Sharpsburg town, Georgia	Turin town, Georgia	Senolia city, Georgia		
Moved in 1989 and earlier	0%	12%	30%	6%	9%	2%		
Moved in 1990 to 1999	10%	10%	24%	15%	8%	9%		
Moved in 2000 to 2009	21%	16%	5%	17%	11%	27%		
Moved in 2010 to 2017	44%	41%	24%	33%	28%	42%		
Moved in 2018 to 2020	18%	14%	18%	28%	37%	17%		
Moved in 2021 or later	7%	7%	0%	1%	7%	3%		

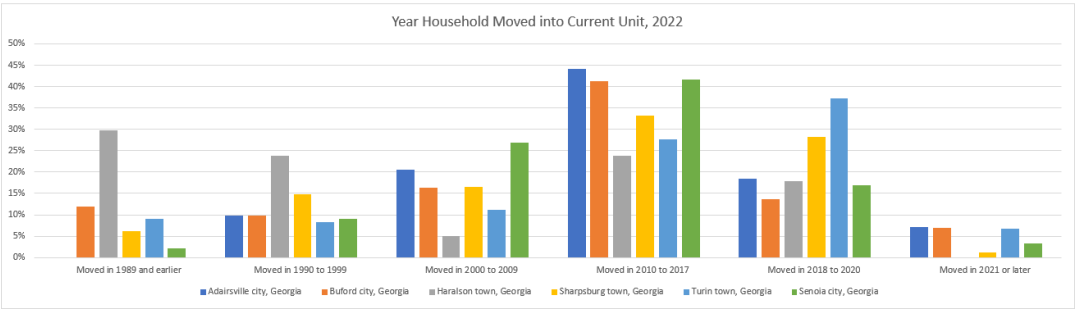


Table:

DP04

Variable(s):

Moved in 1989 and earlier, Moved in 1990 to 1999, Moved in 2000 to 2009, Moved in 2010 to 2017, Moved in 2018 to 2020, Moved in 2021 or later*

Transformation:

All variables divided by Occupied housing units

Two-Period Comparison or Longitudinal:

Neither as this data should only come from the most recent year. Census fucks this up every year so it makes it impossible to look at this data across time because things stop making sense. This is one of two pieces of data that should only be returned for the most recent year no matter what years the user is selecting.

Additional Note:

Over time the Census will add new bins so this table we're generating will need to be updated as Census rolls out new bins for more recent years

Tenure

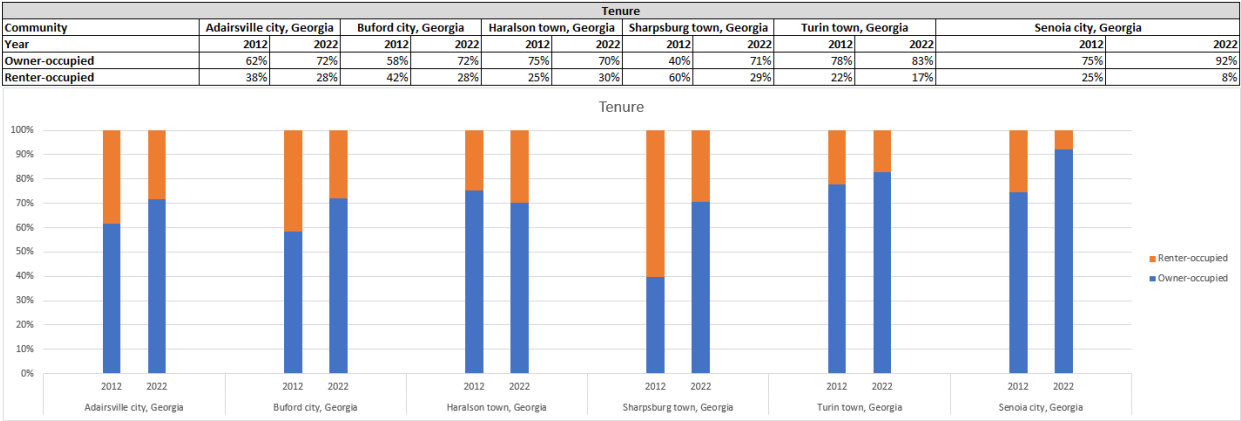


Table:

DP04

Variable(s):

Owner-occupied, Renter-occupied

Transformation:

Both variables divided by Occupied housing units

Two-Period Comparison or Longitudinal:

Two-Period Comparison

Median Home Value

Median Home Value		
Community	2012	2022
Adairsville city, Georgia	\$123,300	\$186,900
Buford city, Georgia	\$169,800	\$292,300
Haralson town, Georgia	\$155,000	\$189,900
Sharpsburg town, Georgia	\$139,700	\$233,600
Turin town, Georgia	\$234,400	\$442,700
Senoia city, Georgia	\$187,300	\$358,400

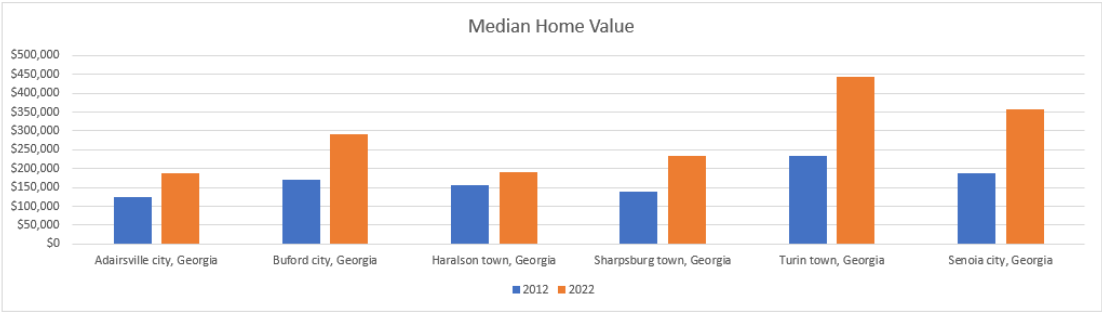


Table:

DP04

Variable(s):

Median (dollars)*

Transformation:

None, direct report

Two-Period Comparison or Longitudinal:

Two-Period Comparison

Additional Note:

Median (dollars) is a variable that is essentially repeated twice in the dataset. The first time it appears it refers to median home value, the second time it appears it refers to median rent paid.

Home Value Distribution

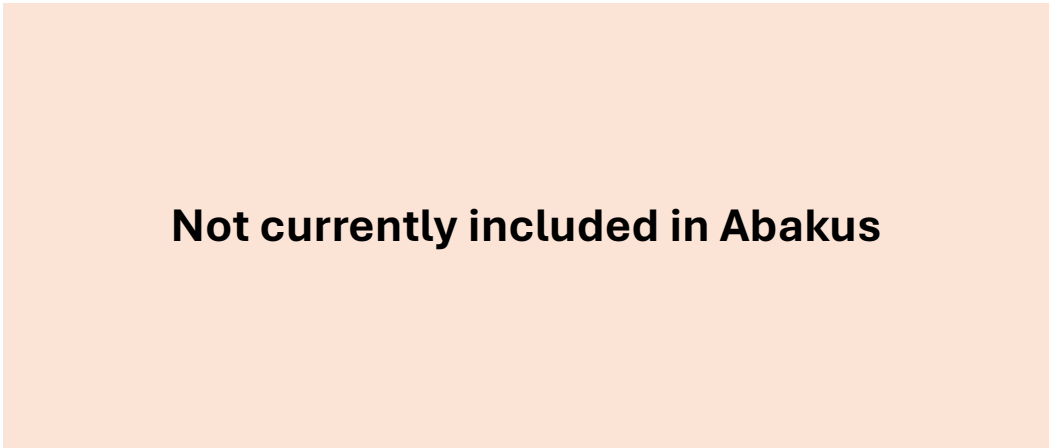


Table:

DP04

Variable(s):

Less than \$50,000; \$50,000 to \$99,999; \$100,000 to \$149,999; \$150,000 to \$199,999; \$200,000 to \$299,999; \$300,000 to \$499,999; \$500,000 to \$999,999; \$1,000,000 or more

Transformation:

None, direct report OR as a percentage by dividing all variables by Owner-occupied

Two-Period Comparison or Longitudinal:

Two-Period Comparison

Median Rent

Median Rent		
Community	2012	2022
Adairsville city, Georgia	\$697	\$1,190
Buford city, Georgia	\$852	\$1,147
Haralson town, Georgia	\$932	\$1,141
Sharpsburg town, Georgia	\$911	\$1,133
Turin town, Georgia	\$905	\$913
Senoia city, Georgia	\$981	\$1,230

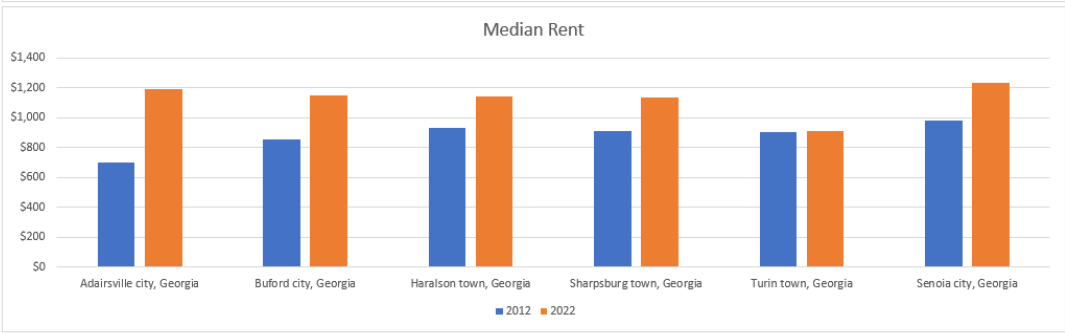


Table:

DP04

Variable(s):

Median (dollars)*

Transformation:

None, direct report

Two-Period Comparison or Longitudinal:

Two-Period Comparison

Additional Note:

Median (dollars) is a variable that is essentially repeated twice in the dataset. The first time it appears it refers to median home value, the second time it appears it refers to median rent paid.

Rent Paid Distribution

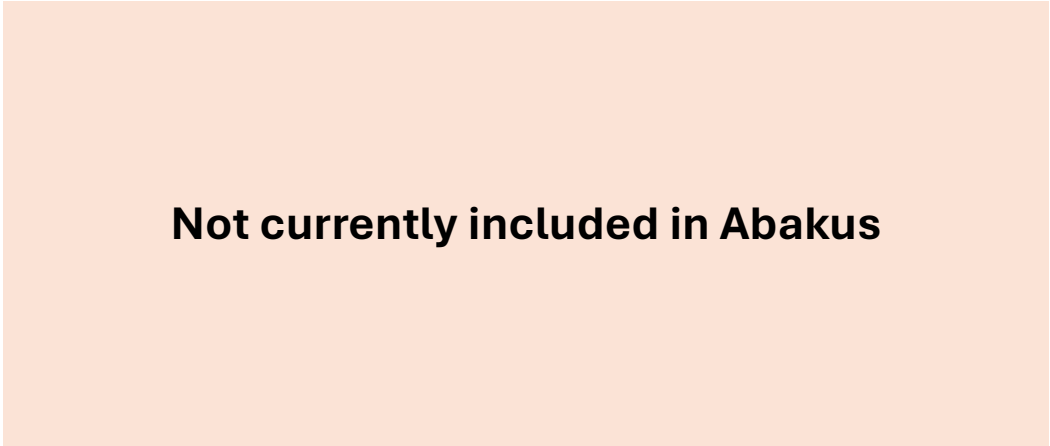


Table:

DP04

Variable(s):

Less than \$500; \$500 to \$999; \$1,000 to \$1,499; \$1,500 to \$1,999; \$2,000 to \$2,499; \$2,500 to \$2,999; \$3,000 or more

Transformation:

None, direct report OR as a percentage by dividing all variables by Renter-occupied

Two-Period Comparison or Longitudinal:

Two-Period Comparison

Owner Cost Burden

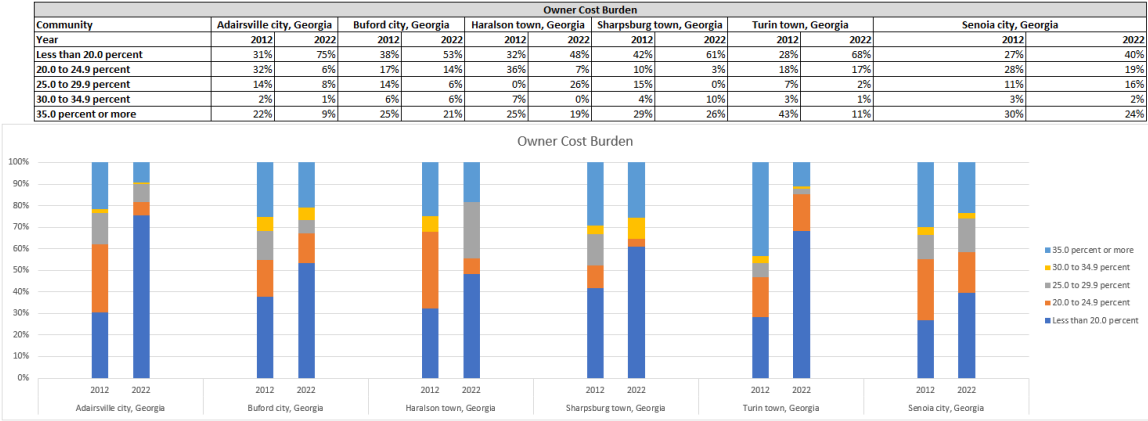


Table:

DP04

Variable(s):

Housing units with a mortgage (excluding units where SMOCAPI cannot be computed), Less than 20.0 percent, 20.0 to 24.9 percent, 25.0 to 29.9 percent, 30.0 to 34.9 percent, 35.0 percent or more

Transformation:

Divide the last five variables by the first variable, Housing units with a mortgage (excluding units where SMOCAPI cannot be computed)

Two-Period Comparison or Longitudinal:

Two-Period Comparison

Renter Cost Burden

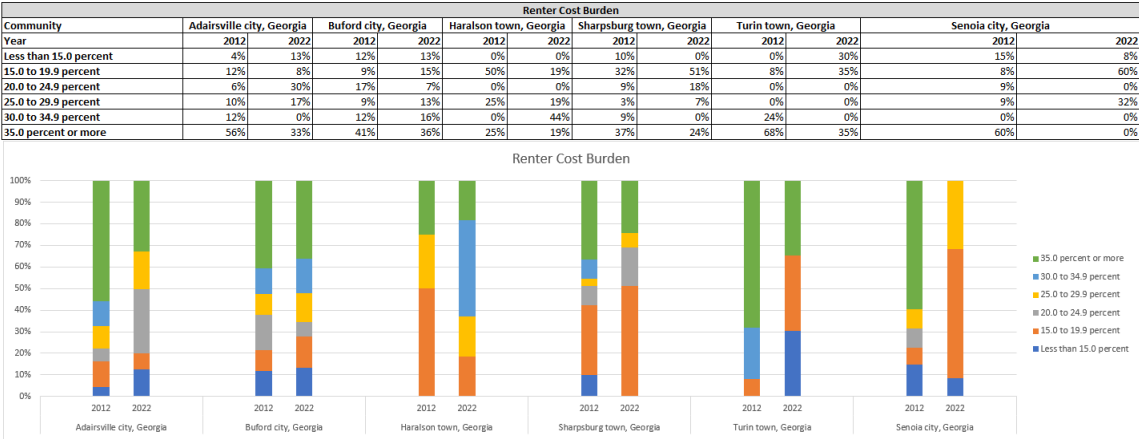


Table:

DP04

Variable(s):

Occupied units paying rent (excluding units where GRAPI cannot be computed), Less than 15.0 percent, 15.0 to 19.9 percent, 20.0 to 24.9 percent, 25.0 to 29.9 percent, 30.0 to 34.9 percent, 35.0 percent or more

Transformation:

Divide the last six variables by the first variable, Housing units paying rent (excluding units where SMOCAPI cannot be computed)

Two-Period Comparison or Longitudinal:

Two-Period Comparison

Age by Cohort

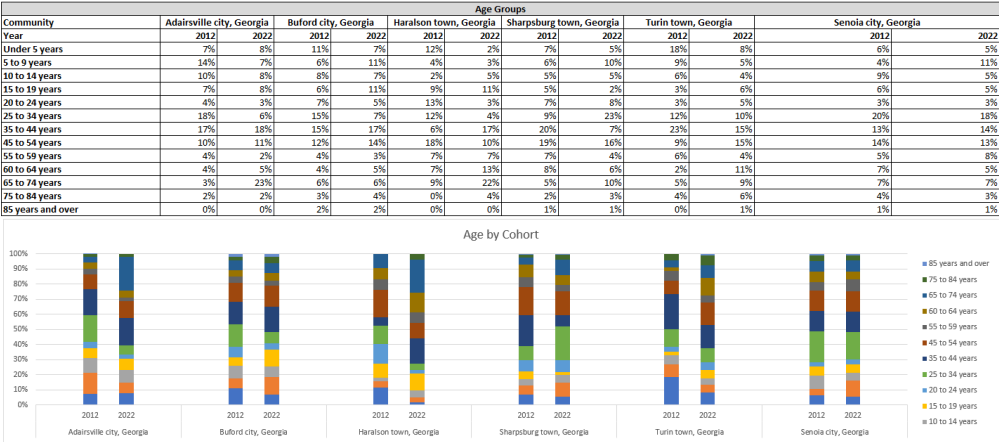


Table:

DP05

Variable(s):

Under 5 years, 5 to 9 years, 10 to 14 years, 15 to 19 years, 20 to 24 years, 25 to 34 years, 35 to 44 years, 45 to 54 years, 55 to 59 years, 60 to 64 years, 65 to 74 years, 75 to 84 years, 85 years and over

Transformation:

Divide all variables by Total population

Two-Period Comparison or Longitudinal:

Two-Period Comparison

Age by Cohort Simplified

Age Groups (Simplified)												
Community	Adairsville city, Georgia		Buford city, Georgia		Haralson town, Georgia		Sharpsburg town, Georgia		Turin town, Georgia		Senoia city, Georgia	
Year	2012	2022	2012	2022	2012	2022	2012	2022	2012	2022	2012	2022
Under 20	37%	31%	31%	36%	27%	21%	22%	22%	35%	23%	26%	27%
20 to 45	39%	27%	37%	29%	31%	23%	37%	37%	38%	30%	36%	35%
45 to 65	18%	18%	21%	22%	32%	30%	33%	27%	18%	31%	26%	27%
65 and Over	6%	25%	11%	13%	9%	26%	7%	14%	9%	16%	12%	12%

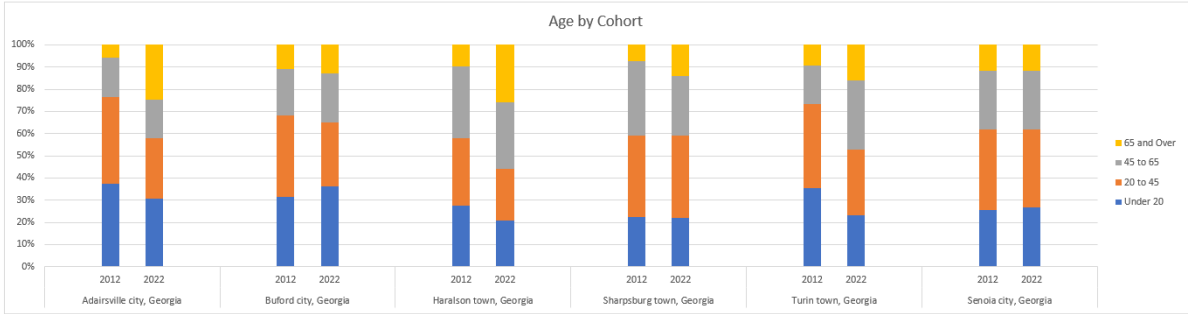


Table:

DP05

Variable(s):

Under 5 years, 5 to 9 years, 10 to 14 years, 15 to 19 years, 20 to 24 years, 25 to 34 years, 35 to 44 years, 45 to 54 years, 55 to 59 years, 60 to 64 years, 65 to 74 years, 75 to 84 years, 85 years and over

Transformation:

Add the percentages to create new groups called Under 20, 20 to 45, 45 to 65 and 65 and Over

Two-Period Comparison or Longitudinal:

Two-Period Comparison

Median Age

Not currently included in Abakus

Table:

DP05

Variable(s):

Median age (years)

Transformation:

None, direct report

Two-Period Comparison or Longitudinal:

Two-Period Comparison

Race

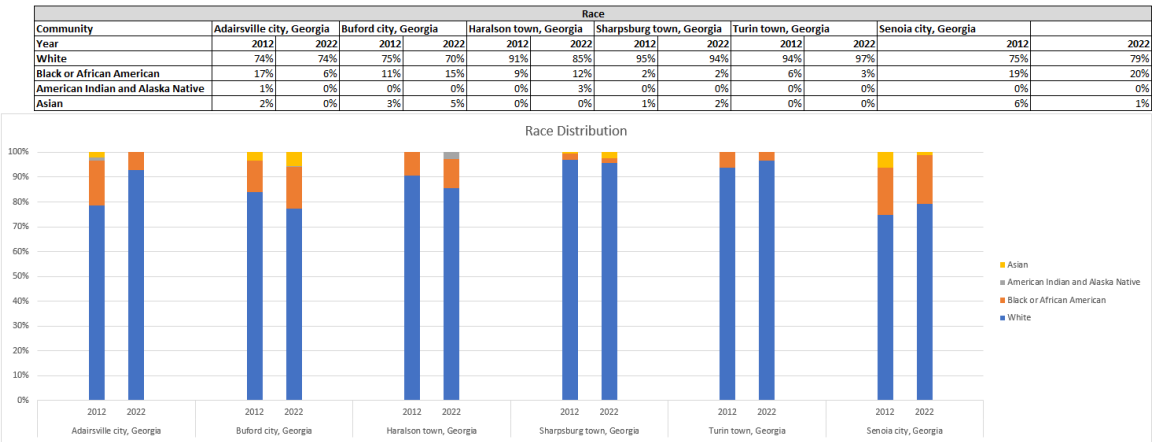


Table:

DP05

Variable(s):

One race, White, Black or African American, American Indian and Alaska Native, Asian, Native Hawaiian and Other Pacific Islander, Some Other Race

Transformation:

Divide each of the six racial categories by One race

Two-Period Comparison or Longitudinal:

Two-Period Comparison

Hispanic Ethnicity

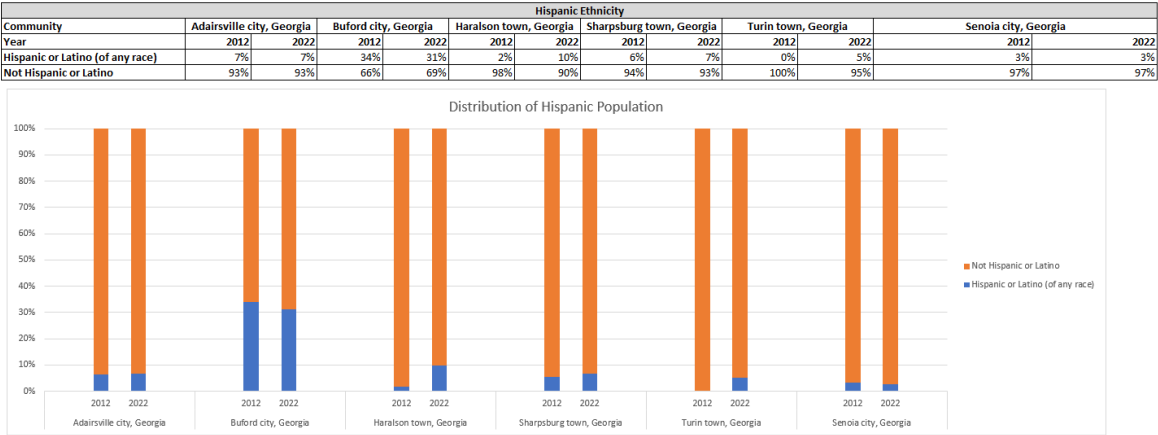


Table:

DP05

Variable(s):

Hispanic or Latino (of any race), Not Hispanic or Latino

Transformation:

Divide both variables by Total population

Two-Period Comparison or Longitudinal:

Two-Period Comparison

Household Income

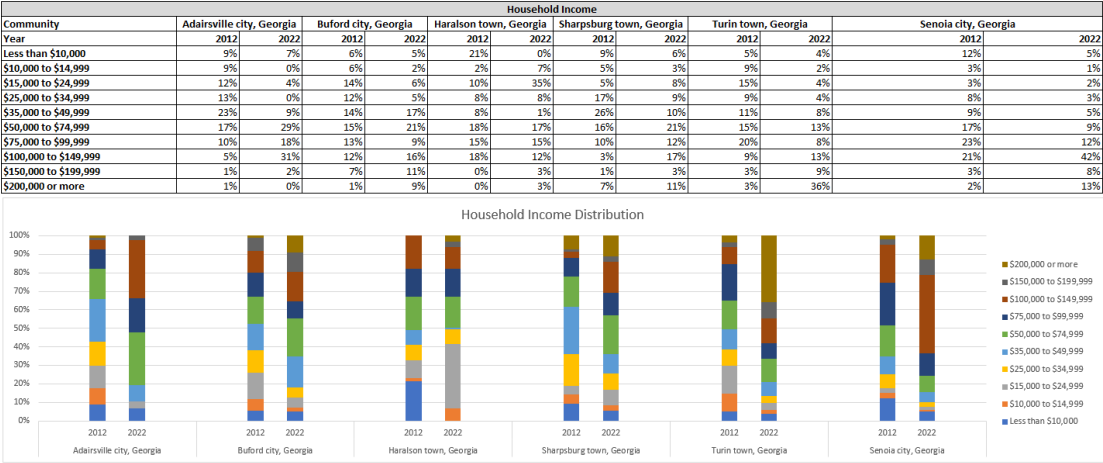


Table:

S1901

Variable(s):

Total, Less than \$10,000, \$10,000 to \$14,999, \$15,000 to \$24,999, \$25,000 to \$34,999, \$35,000 to \$49,999, \$50,000 to \$74,999, \$75,000 to \$99,999, \$100,000 to \$149,999, \$150,000 to \$199,999, \$200,000 or more

Transformation:

Divide all income descriptive variables by Total

Two-Period Comparison or Longitudinal:

Two-Period Comparison

Household Income Simplified

munity	Household Income (Simplified)											
	Adairsville city, Georgia		Buford city, Georgia		Haralson town, Georgia		Sharpsburg town, Georgia		Turin town, Georgia		Senoia city, Georgia	
	2012	2022	2012	2022	2012	2022	2012	2022	2012	2022	2012	2022
han \$35,000	43%	11%	38%	18%	41%	50%	36%	26%	38%	13%	25%	
00 to \$49,000	23%	9%	14%	8%	17%	8%	26%	10%	11%	8%	9%	
00 to \$74,999	17%	29%	15%	21%	18%	17%	16%	21%	15%	13%	17%	
00 to \$99,999	10%	18%	13%	9%	15%	15%	10%	12%	20%	8%	23%	
000 to \$149,999	5%	31%	12%	16%	18%	12%	3%	17%	9%	13%	21%	
000 or more	2%	2%	8%	20%	0%	6%	9%	14%	6%	45%	5%	

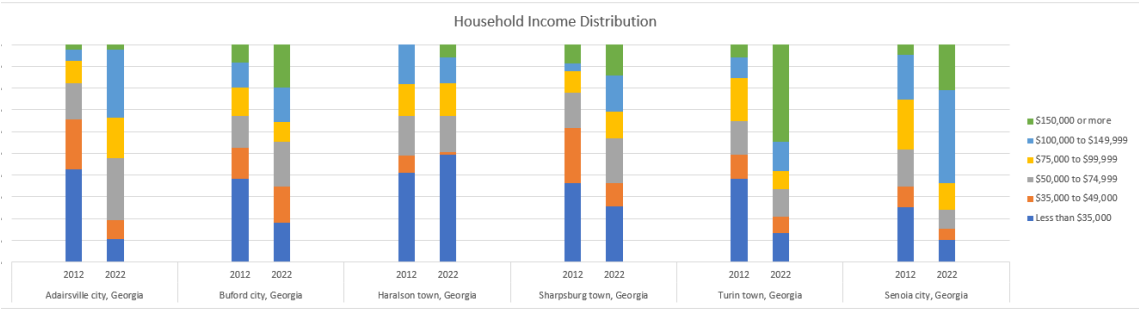


Table:

S1901

Variable(s):

Total, Less than \$10,000, \$10,000 to \$14,999, \$15,000 to \$24,999, \$25,000 to \$34,999, \$35,000 to \$49,999, \$50,000 to \$74,999, \$75,000 to \$99,999, \$100,000 to \$149,999, \$150,000 to \$199,999, \$200,000 or more

Transformation:

Add the percentages to create six new bins: Less than \$35,000, \$35,000 to \$50,000, \$50,000 to \$75,000, \$75,000 to \$100,000, \$100,000 to \$150,000, and \$150,000 or more

Two-Period Comparison or Longitudinal:

Two-Period Comparison

Median Household Income

Median Household Income		
Community	2012	2022
Adairsville city, Georgia	\$40,565	\$90,062
Buford city, Georgia	\$46,793	\$68,165
Haralson town, Georgia	\$55,417	-
Sharpsburg town, Georgia	\$43,750	\$69,531
Turin town, Georgia	\$50,625	\$139,167
Senoia city, Georgia	\$62,350	\$114,423

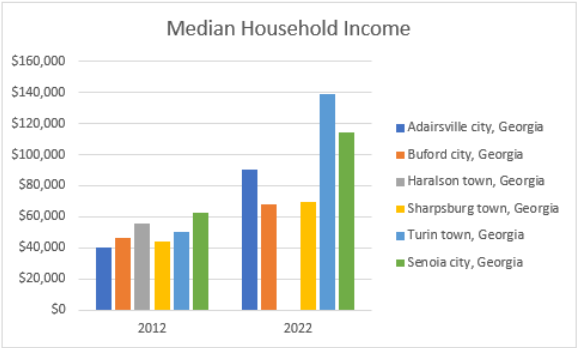


Table:

S1901

Variable(s):

Median income (dollars)

Transformation:

None, direct report

Two-Period Comparison or Longitudinal:

Two-Period Comparison

Tenure by Age

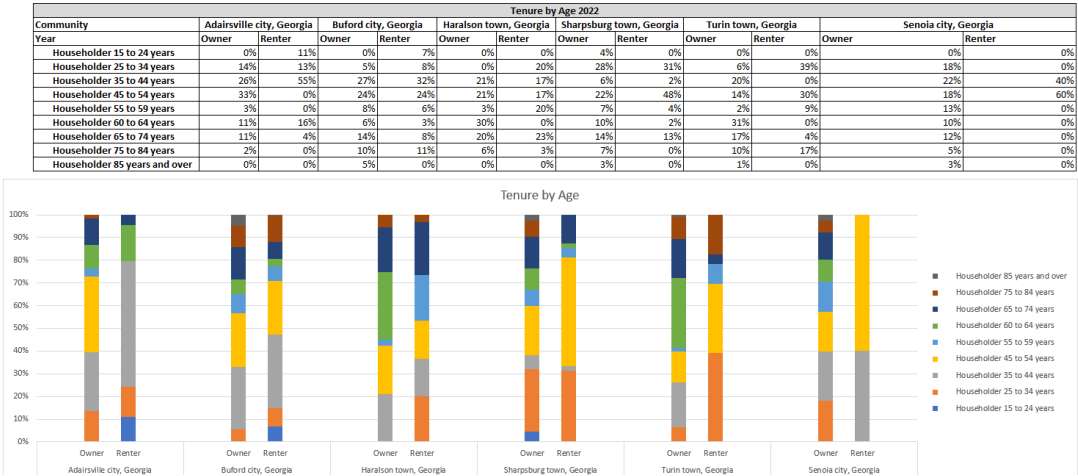


Table:

B25007

Variable(s):

Owner occupied:, Householder 15 to 24 years, Householder 25 to 34 years, Householder 35 to 44 years, Householder 45 to 54 years, Householder 55 to 59 years, Householder 60 to 64 years, Householder 65 to 74 years, Householder 75 to 84 years, Householder 85 years and over

Renter occupied:, Householder 15 to 24 years, Householder 25 to 34 years, Householder 35 to 44 years, Householder 45 to 54 years, Householder 55 to 59 years, Householder 60 to 64 years, Householder 65 to 74 years, Householder 75 to 84 years, Householder 85 years and over

Transformation:

Divide the householder age groups by either Owner occupied: or Renter occupied: depending on what section of the table they fall under

Two-Period Comparison or Longitudinal:

Two-Period Comparison

Tenure by Income

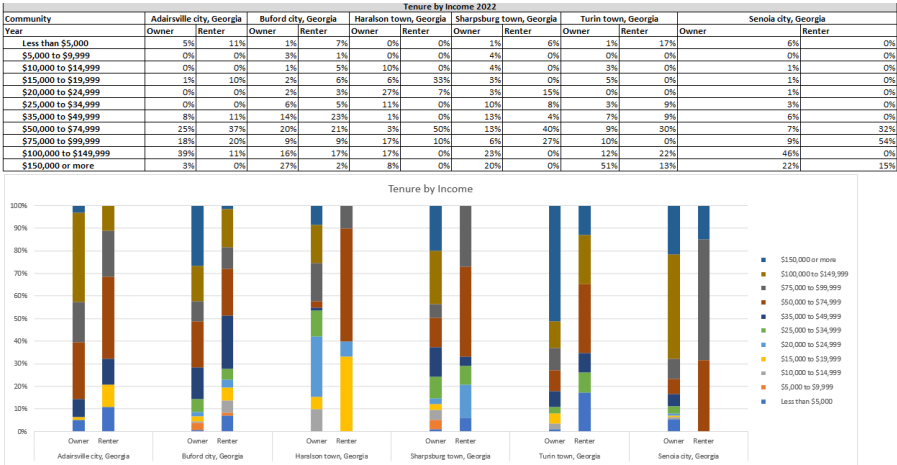


Table:

B25118

Variable(s):

Owner occupied:, Less than \$5,000, \$5,000 to \$9,999, \$10,000 to \$14,999, \$15,000 to \$19,999, \$20,000 to \$24,999, \$25,000 to \$34,999, \$35,000 to \$49,999, \$50,000 to \$74,999, \$75,000 to \$99,999, \$100,000 to \$149,999, \$150,000 or more

Renter occupied:, Less than \$5,000, \$5,000 to \$9,999, \$10,000 to \$14,999, \$15,000 to \$19,999, \$20,000 to \$24,999, \$25,000 to \$34,999, \$35,000 to \$49,999, \$50,000 to \$74,999, \$75,000 to \$99,999, \$100,000 to \$149,999, \$150,000 or more

Transformation:

Divide the householder income groups by either Owner occupied: or Renter occupied: depending on what section of the table they fall under

Two-Period Comparison or Longitudinal:

Two-Period Comparison

Household Size

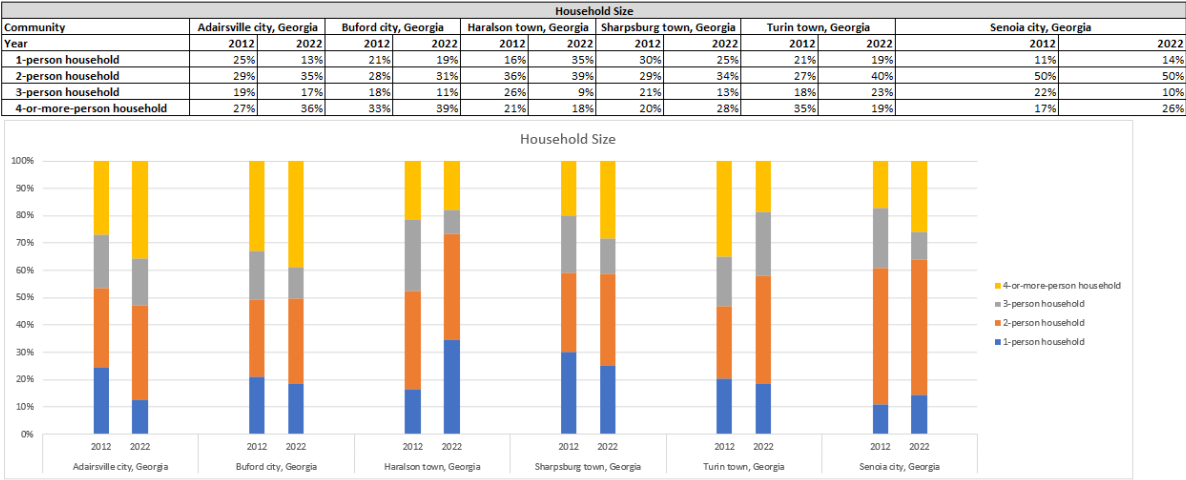


Table:

S2501

Variable(s):

Occupied housing units, 1-person household, 2-person household, 3-person household, 4-or-more-person household

Transformation:

For 2017-Present: Divide the household size figures by Occupied housing units

For 2016 and Earlier: Multiply the household size percentages by Occupied housing units

Two-Period Comparison or Longitudinal:

Two-Period Comparison

Household Type

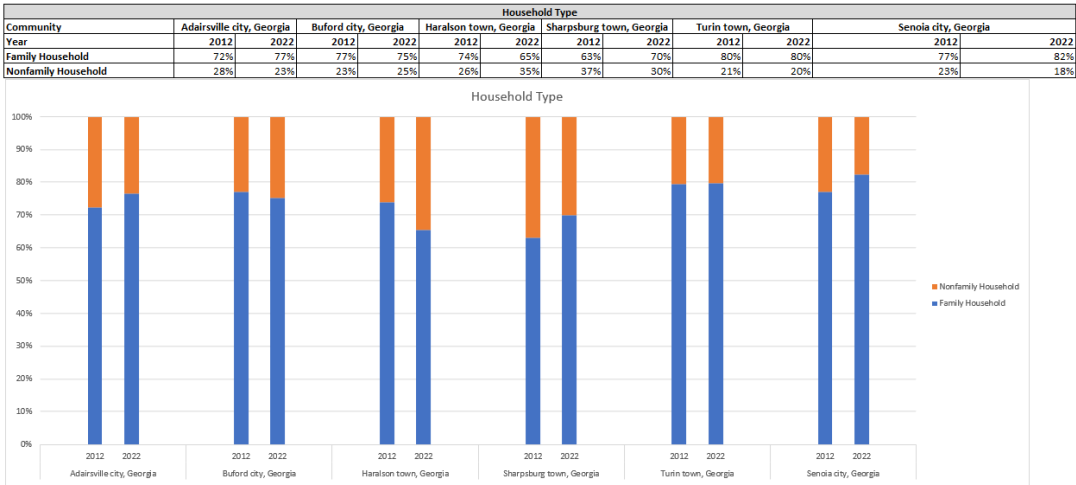


Table:

S2501

Variable(s):

Occupied housing units, Family households, Nonfamily households

Transformation:

For 2017-Present: Divide the household size figures by Occupied housing units

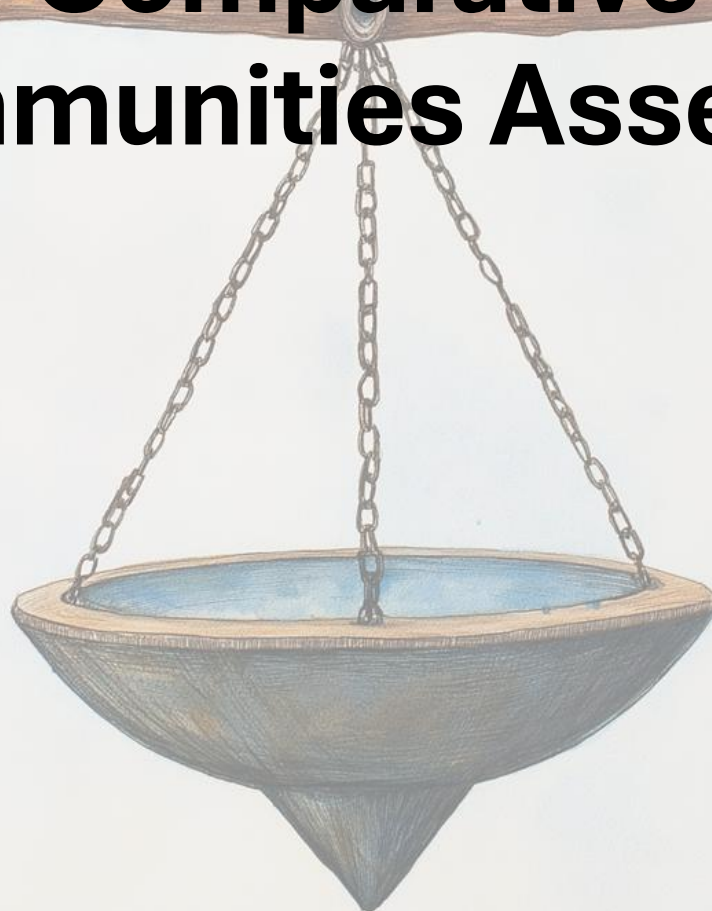
For 2016 and Earlier: Multiply the household size percentages by Occupied housing units

Two-Period Comparison or Longitudinal:

Two-Period Comparison



Comparative Communities Assessor



Comparative Communities Assessor

Places Most Similar to:			
Place	Housing Units 2022	Households 2022	Median Household Income 2021
Adairsville city, Georgia	1,771	1,414	\$75,964

Selected Comparative Communities:				
Rank	Place	Housing Units 2022	Households 2022	Median Household Income 2021
2,055	Buford city, Georgia	5,800	5,019	\$54,321
182	Haralson town, Georgia	105	101	-
1,208	Sharpsburg town, Georgia	166	163	\$67,763
600	Turin town, Georgia	144	134	\$96,250
656	Senoia city, Georgia	1,892	1,796	\$99,120

*Rankings are out of 3,978 Census Designated Places and 534 Counties

State Filter -> All States

Rank	Place	Housing Units 2022	Households 2022	Median Household Income 2021
1	Holden Lakes CDP, Florida	1,793	1,385	
2	Pleasant View city, Tennessee	2,052	1,950	\$90,191
3	Kure Beach town, North Carolina	2,199	973	\$89,828
4	Cudjoe Key CDP, Florida	1,653	962	\$89,556
5	Pine Knoll Shores town, North Carolina	2,167	691	\$89,861
6	Burnt Store Marina CDP, Florida	1,995	1,210	\$90,893
7	Belle Meade city, Tennessee	979	915	250,000+
8	Fairfield Plantation CDP, Georgia	2,292	2,162	\$89,762

Tables:

DP04, S1101, S1901

Variable(s):

Total housing units, Total households, Median income (dollars)

Details:

The comparative communities assessor runs by looking at three variables for the community of primary interest and using a sum of squared difference method to approximate the communities that are most similar to the community of primary interest. The smaller the sum of squared difference is for every other community, the higher ranking it receives. Within the current Abakus tool, number 1 is always the community of primary interest so adjustments have been made to display number 2 as number 1 and so on.

Comparative Communities Assessor

City Long	State	Housing Units	Households	HHI	Housing Squared Difference	Population Squared Difference	HHI Squared Difference	Sum of Squared Difference	Rank
Abanda CDP, Alabama	Alabama	79	79	\$25,787	2862864	1782225	4131275625	4135920714	4565
Abbeville city, Alabama	Alabama	1,303	978	\$36,944	219024	190096	2821521924	2821931044	3719
Adamsville city, Alabama	Alabama	1,757	1,577	\$46,150	196	26569	1928263744	1928290509	2757
Addison town, Alabama	Alabama	361	277	\$44,000	1988100	1292769	2121707844	2124988713	2980
Akron town, Alabama	Alabama	173	86	\$28,438	2553604	1763584	3797517376	3801834564	4424
Alabaster city, Alabama	Alabama	12,459	11,976	\$83,181	114233344	111555844	47348161	273137349	941
Albertville city, Alabama	Alabama	8,267	7,576	\$51,718	42198016	37970244	1470262336	1550430596	2353
Alexander City city, Alabama	Alabama	7,211	5,978	\$41,778	29593600	20830096	2331344656	2381768352	3259
Alexandria CDP, Alabama	Alabama	1,579	1,317	\$88,090	36864	9409	3888784	3935057	86
Aliceville city, Alabama	Alabama	993	894	\$30,411	605284	270400	3558241801	3559117485	4300
Allgood town, Alabama	Alabama	231	211	\$56,250	2371600	1447209	1143251344	1147070153	1870
Altoona town, Alabama	Alabama	568	458	\$40,000	1447209	913936	2506203844	2508564989	3403
Andalusia city, Alabama	Alabama	4,147	3,097	\$40,516	5645376	2832489	2454806116	2463283981	3349
Anderson town, Alabama	Alabama	112	111	\$76,250	2752281	1697809	190771344	195221434	840
Anniston city, Alabama	Alabama	11,705	9,473	\$39,928	98684356	64947481	2513417956	2677049793	3569
Arab city, Alabama	Alabama	3,635	3,336	\$50,321	3474496	3694084	1579347081	1586515661	2391
Ardmore town, Alabama	Alabama	566	552	\$37,448	1452025	743044	2768232996	2770428065	3675
Argo town, Alabama	Alabama	1,676	1,537	\$80,733	9025	15129	87030241	87054395	666
Ariton town, Alabama	Alabama	292	248	\$31,406	2187441	1359556	3440526336	3444073333	4236
Arley town, Alabama	Alabama	196	171	\$44,167	2480625	1545049	2106351025	2110376699	2965
Ashford town, Alabama	Alabama	1,011	859	\$47,167	577600	308025	1839981025	1840866650	2653
Ashland town, Alabama	Alabama	829	709	\$25,458	887364	497025	4173676816	4175061205	4579
Ashville city, Alabama	Alabama	910	777	\$42,875	741321	405769	2226612969	2227760059	3119
Athens city, Alabama	Alabama	11,586	10,441	\$52,211	96334225	81486729	1432698201	1610519155	2421
Atmore city, Alabama	Alabama	3,438	2,725	\$29,104	2778889	1718721	3715877764	3720375374	4376
Attalla city, Alabama	Alabama	3,107	2,568	\$46,278	1784896	1331716	1917038656	1920155268	2728

Details:

Above is a screenshot of the back-end of the assessor within Excel. In the current tool, if the user selects Place, all places are considered, if the user selects County, all counties are considered instead.

The current tool’s geographic limitations may actually be a positive when it comes to this module, as all of the communities it can return are within the same region of the United States and would therefore have greater relevance for comparison than communities that are further away.

As we’re working through this, maybe we should consider designating regions and having every state assigned a region so that only communities within the states within the region that the community is within are considered in this analysis.

Squared Difference:

$$(a - b)^2 = a^2 - 2ab + b^2$$

Sum of Squared Difference:

$$SSD = \sum_{i=1}^n (a_i - b_i)^2$$

Comparative Communities Assessor

1 Adairsville Georgia	Rank	Place	Housing Units 2022	Households 2022	Median Household Income 2021
2 Holden Lal Florida	1	Holden Lakes CDP, Florida	1,793	1,385	
3 Pleasant V Tennessee	2	Pleasant View city, Tennessee	2,052	1,950	\$90,191
4 Kure Beacl North Carolina	3	Kure Beach town, North Carolina	2,199	973	\$89,828
5 Cudjoe Key Florida	4	Cudjoe Key CDP, Florida	1,653	962	\$89,556
6 Pine Knoll North Carolina	5	Pine Knoll Shores town, North Carolina	2,167	691	\$89,861
7 Burnt Stor Florida	6	Burnt Store Marina CDP, Florida	1,995	1,210	\$90,893
8 Belle Mead Tennessee	7	Belle Meade city, Tennessee	979	915	250,000+
9 Fairfield Pl Georgia	8	Fairfield Plantation CDP, Georgia	2,292	2,162	\$89,762
10 Utica CDP, South Carolina	9	Utica CDP, South Carolina	971	782	-
11 Irwin CDP, South Carolina	10	Irwin CDP, South Carolina	750	702	-
12 Isle of Ho Georgia	11	Isle of Hope CDP, Georgia	1,011	961	\$90,944
13 Pegram to Tennessee	12	Pegram town, Tennessee	912	848	\$89,293
14 Alamo tow Georgia	13	Alamo town, Georgia	722	664	-
15 Pimlico CD South Carolina	14	Pimlico CDP, South Carolina	681	681	-
16 Tangerine Florida	15	Tangerine CDP, Florida	1,347	1,265	\$91,324
17 Watts Mill South Carolina	16	Watts Mills CDP, South Carolina	660	567	-
18 Matlacha t Florida	17	Matlacha CDP, Florida	719	437	-
19 Pinecraft t Florida	18	Pinecraft CDP, Florida	874	277	-
20 Weeki Wa Florida	19	Weeki Wachee Gardens CDP, Florida	798	340	-
21 Palm Shor Florida	20	Palm Shores town, Florida	598	569	\$90,221
22 Black Ham Florida	21	Black Hammock CDP, Florida	570	570	-
23 Olympia Cl South Carolina	22	Olympia CDP, South Carolina	611	512	-

Details:

Above is a screenshot of the front-end, which pulls in only the top 101 communities from the back-end of the assessor. (Note that in the back-end I’ve used the rank formula to assign 1 to the lowest number)

This then removes the first community, as it is the community of primary interest and then reports out the Housing Units, Households and Median Household Income for these communities.

One important consideration that I realized after building out this tool is that using the most current year for Median Household Income is not always best actually, as there’s sometimes a lag in reporting, making the dataset less accurate. As we are updating into 2023, only now would I say we should use 2022 Median Household Income.

A watercolor illustration of a residential construction site. In the foreground, a dirt road with tire tracks leads into the scene. On the left, several multi-story houses are under construction, showing wooden framing and some siding. On the right, another house is partially visible. In the background, a row of completed, two-story houses with dark roofs and light-colored siding stands. The sky is blue with soft, white clouds. The overall style is artistic and painterly.

Housing Demand Projections

Housing Demand Projections

Year	10-Year Projection	Households	10-Year Projection
	Population		Average Household Size
2010	4,404	1,527	2.88
2011	4,520	1,623	2.78
2012	4,605	1,539	2.99
2013	4,660	1,490	3.13
2014	4,692	1,501	3.13
2015	4,712	1,524	3.09
2016	4,740	1,665	2.85
2017	4,756	1,765	2.69
2018	4,792	1,838	2.61
2019	4,856	1,830	2.65
2020	4,926	1,467	3.36
2021	4,856	1,371	3.54
2022	4,911	1,414	3.47
2023	4,943	1,402	3.53
2024	4,975	1,390	3.58
2025	5,007	1,379	3.63
2026	5,039	1,367	3.69
2027	5,072	1,355	3.74
2028	5,104	1,344	3.80
2029	5,137	1,333	3.86
2030	5,170	1,321	3.91
2031	5,204	1,310	3.97
2032	5,237	1,299	4.03
2033	5,271	1,288	4.09
2034	5,305	1,277	4.15

Tables:

DP05, S1101

Variable(s):

Total population, Total households, Average household size

Details:

All housing demand runs off of one key thing: household growth. The table above populates population, households and average household size for every year that there is data.

Based on this data, 5-Year and 10-Year historic growth rates are calculated (using RATE() formulas in Excel) which the user can then select to project forward population and household size.

We don't project households directly because we're trying to also take into consideration whether we're getting more small households or more large households by projecting household size forward and then applying that to population. Households are calculated by dividing population by average household size.

Details:

The turnover rates listed below are never seen by the user, but the user is able to select which turnover rate to apply, which in turn informs the total housing demand across the period.

These turnover percentages are examples of “Expert Assumptions” that I have programmed into the tool.

Housing Demand Projections

Insert Population Custom Rate ->

2.10%

Insert Household Size Custom Rate ->

0.64%

	2024	2034	2024 - 2034 Change
Population	4,975	5,305	331
Households	1,390	1,277	(113)

10-Year Housing Demand	
10-Year Total	34
10-Year Annual	3

Select Internal Turnover Rate ->

Aggressive Internal Turnover

Insert Custom Internal Turnover Rate ->

2.5000%

Details:

The table above represents a few different drop down options that users can select, as well as a few different opportunities for manual entry of growth and turnover rates. If the user changes the years, the housing demand range changes. For example, if 2029 were selected instead of 2034, the box would read 5-Year Housing Demand.

The 10-Year Housing Demand represents the top line output of this module and is given as a period total and an annual number. These numbers are arrived at by subtracting the households present in 2024 from the households present in 2034 (or whichever years selected). This number is then added to the internal turnover numbers identified on the previous page for each of the years in the range.

Housing Demand Projections

10-Year Growth Rate	Distribution of Annual Housing Demand by Age and Income			
	Householder under 25 years	Householder 25 to 44 years	Householder 45 to 64 years	Householder 65 years and older
Less than \$35,000	0	0	0	0
\$35,000 to \$49,999	-	0	0	0
\$50,000 to \$74,999	-	0	0	0
\$75,000 to \$99,999	-	0	0	-
\$100,000 to \$149,999	-	3	0	0
\$150,000 or more	-	-	0	-

Percent Distribution of Annual Housing Demand by Age and Income				
	Householder under 25 years	Householder 25 to 44 years	Householder 45 to 64 years	Householder 65 years and older
Less than \$35,000	0.4%	0.3%	0.4%	0.2%
\$35,000 to \$49,999	0.0%	0.6%	0.2%	0.3%
\$50,000 to \$74,999	0.0%	1.1%	1.7%	0.7%
\$75,000 to \$99,999	0.0%	0.9%	1.3%	0.0%
\$100,000 to \$149,999	0.0%	90.6%	0.8%	0.1%
\$150,000 or more	0.0%	0.0%	0.3%	0.0%

Tables:

B19037

Variable(s):

All Variables

Details:

The table above is arrived at through an intense process of data organization. Census table B19037 is organized in one column, but it needs to be in four columns, with each of the whole numbers given for the income ranges divided by the age group total. This needs to be done for every year there is data. Based on these tables that hold percentages, we then have to calculate the rate of change from one year to another for each cell in the table. This allows us to generate 5-Year and 10-Year historic growth rates that can be applied by the user to estimate the percentages for future.

Every percentage value in the forecasted table is then multiplied by either the total or the annual housing demand to determine housing demand by age and income. This helps us determine what kinds of housing products are going to be demanded in the years to come.